EXECUTIVE REFERENCE DATA
NATIONAL ENERGY MANAGEMENT

NATIONAL ENERGY COUNCIL
2014

PREFACE

In order to provide updated information on energy situations in Indonesia, the National Energy Council deemed necessary to re-publish an improved publication of the 2013 booklet on Executive Reference Data: National Energy Management. This rst edition in 2014 presenting information on energy management up to 2013, which includes socio-economic, energy indicator, the management each type of energy covering oil and gas, coal, renewable energy, and electricity. Slightly di erent from the 2013 edition, this publication provides also information related to energy by sector as well as Indonesia positions in the global context.

The economic growth of Indonesia at an average annual rate of 5.78% during the period 2012 through 2013, has trigerred the increase of domestic energy demand by an average of 4.4% per year. By 2030 the energy demand is estimated to be three times the demand level in 2010, while the increase in energy production in 2030 is forecast to rise at twice that of the year 2010. This condition requires a good energy planning and management to ensure continuity of supply of energy in the long term.

We would like to thank and appreciate all parties, for their contribution to the preparation of this booklet. It is hoped that this publication will be of great bene ts for energy management in Indonesia and a useful tool to the national and global energy discussions.

Jakarta, July 2014

Secretary General of

National Energy Council

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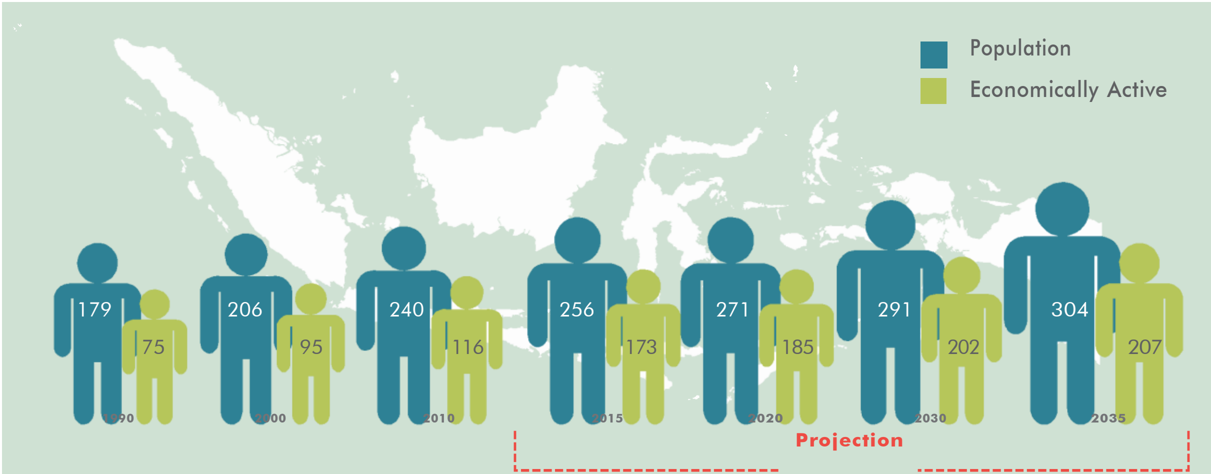
SOCIO - ECONOMIC

POPULATION

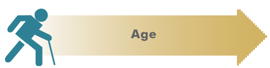
(Million Peoples)



Source: Indonesian Central Bureau of Statistics, reprocessed by NEC

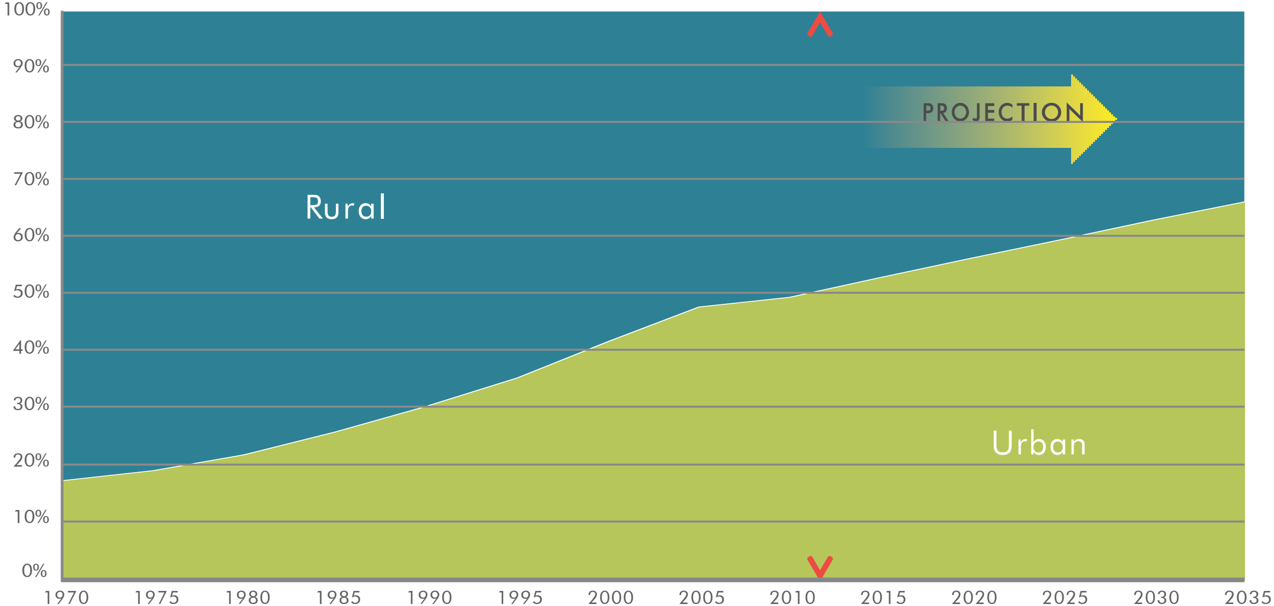
Indonesia will be facing "demographical bonus", in which productive age (age range 15-54 years) is greater 68% compared to the non-productive age This of course would be benecial if there is an increase in employment and the improvement of education in Indonesia.

Education is the most important for school-age children



After experiencing a demographic bonus, Indonesia is expected to experience over-aging, where the number of non-productive age population is much greater.

POPULATION IN RURAL AND URBAN AREAS



Source: Indonesian Central Bureau of Statistics, reprocessed by NEC

In 1970, number of population in urban area 17.5% increase 48.1% in 2005 (increase an average 2.9 % per year).
In period 2005 - 2013, Urban people increase slowly by an average of 1.2 % per year.

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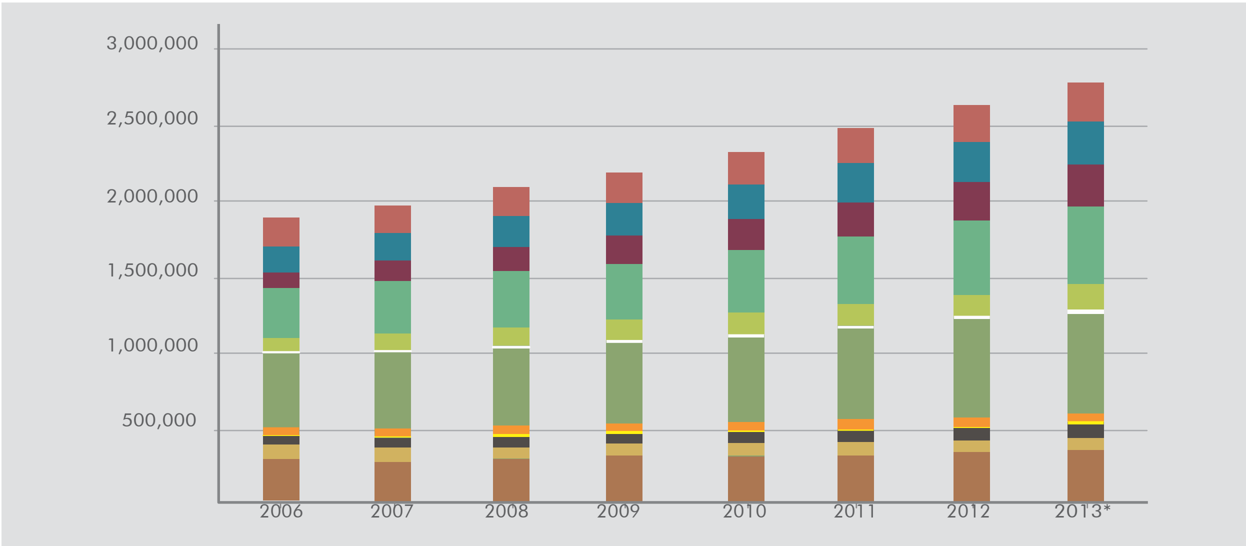
In 2035, the population in the urban areas is estimated/projected to be increased by the composition of 66.6% living in town/city and 33.4% living in villages. The change of a village to become a town has caused the increase in number and density of population, and the economic activity is no longer relies on the agricultural sector, due to the improvement of infrastructure.

GDP PER SECTOR (CONSTANT 2000)

(Billion IDR)



Source: Indonesian Central Bureau of Statistics, reprocessed by NEC

- Services
- Transport & Communication
- Construction
- Non-Oil & Gas Manf. Industry
- Quarrying
- Oil & Gas Mining
- Finance
- Trade, Hotel & Restaurants
- Electricity, Gas & Water Supply
- Oil & Gas Manf. Industry
- Non-Oil & Gas Mining
- Agriculture

* Temporary Value

In 2013, GDP of Indonesia is 2,770 trillion rupiahs or increased one and a half times when compared with the year 2006 (growth 5.96 % per year), with the growth occurs in all sectors (the highest growth is in the in transportation and communication sectors about 12.9 % per year). Oil & Gas and Mining Industry decreased about 1.1 % and 0.99 % per year.

In 2013, the share of mining, oil and gas Industry to the Indonesia's total GDP is about 3% and 2% respectively.

GDP BY REGION

(GDPR in Billion IDR)

Source:BPS - Statistics Indonesia, reprocessed by NEC

400,001 – 500,000

> 600,000

300,001 – 400,000

500,001 – 600,000

200,001 – 300,000

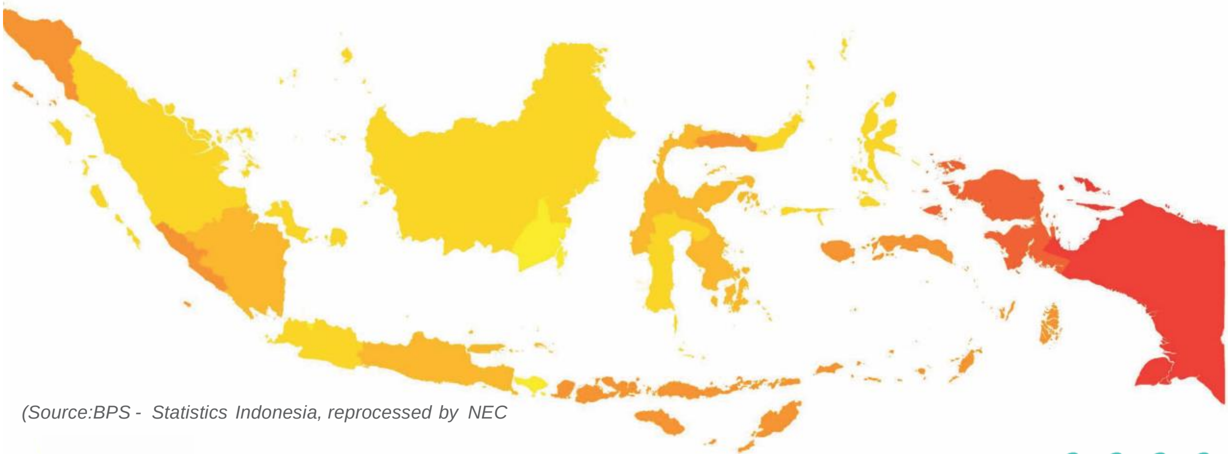
100,001 – 200,000

50,000 – 100,000

< 50,000

POOR PEOPLE, 2013

(%)



(Source:BPS - Statistics Indonesia, reprocessed by NEC

<5%

5 - 10 %

11 - 15 %

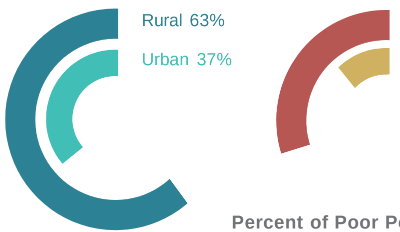
16 - 20 %

20 - 25 %

26 - 30 %

>30 %

*) September 2013



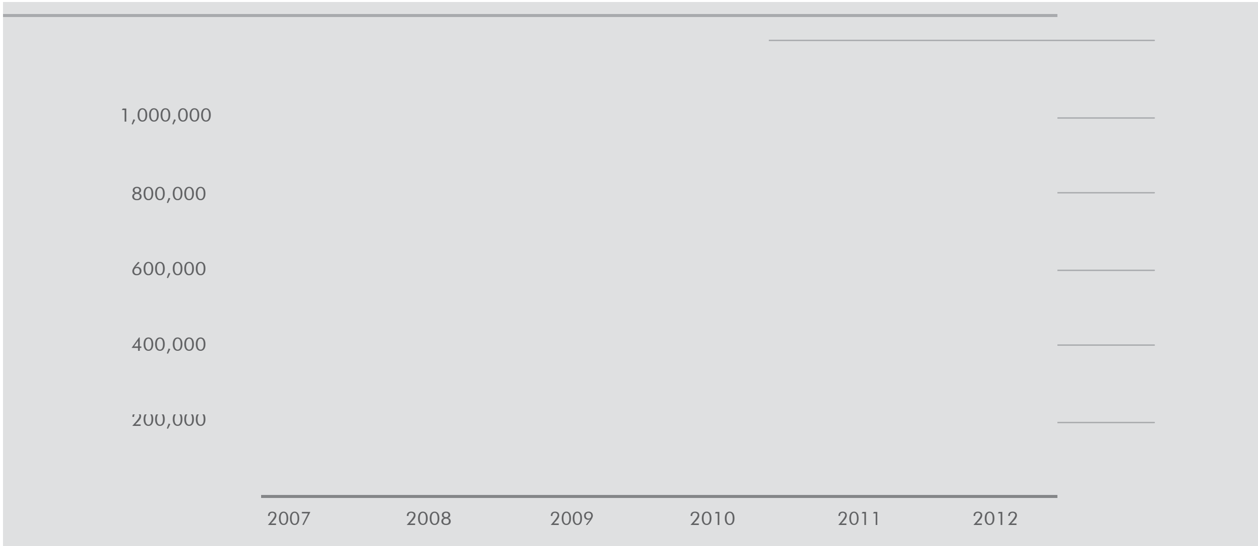
Percent of Poor Population

28 million people (11%) of Indonesia's population is poor, of which 8.53% domiciling in the urban area and 14:42% residing in the rural area.

ENERGY INDICATOR

PRIMARY ENERGY SUPPLY

(MBOE)



	2007	2008	2009	2010	2011	2012
Others RE	2.873	3.193	3.991	4.434	4.668	6.025
Hydro	28,451	29,060	28,696	44,003	30,074	31,795
Gas	183,624	193,352	253,198	269,942	279,286	275,950
Oil	474,033	464,704	546,050	572,249	497,729	485,045
Coal	258,174	205,492	236,439	281,400	334,143	344,400

Source: Ministry of Energy and Mineral Resources/MEMR, reprocessed by NEC
**) Excluded Biomass*

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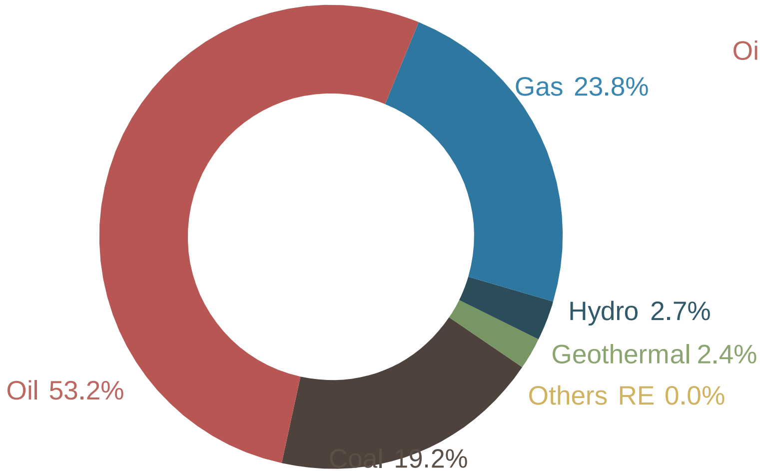
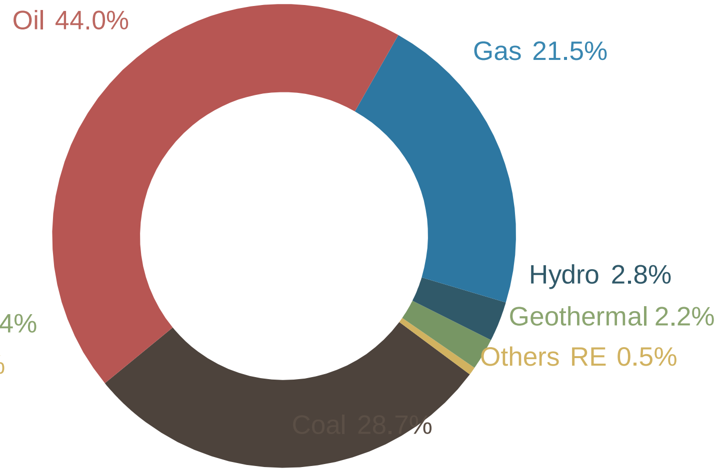
PRIMARY ENERGY MIX

2003
858 Million BOE

3.8% p.a.
Annual growth

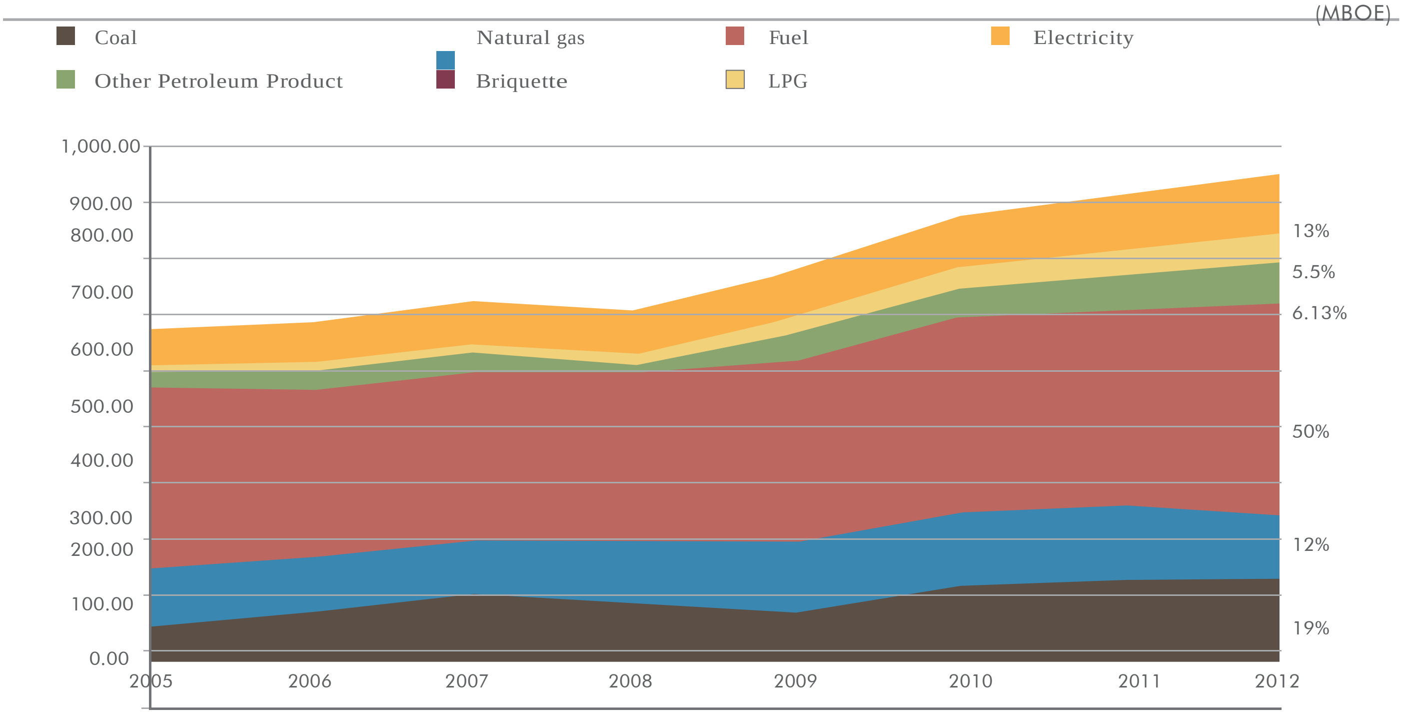
2013*
1,243 Million BOE





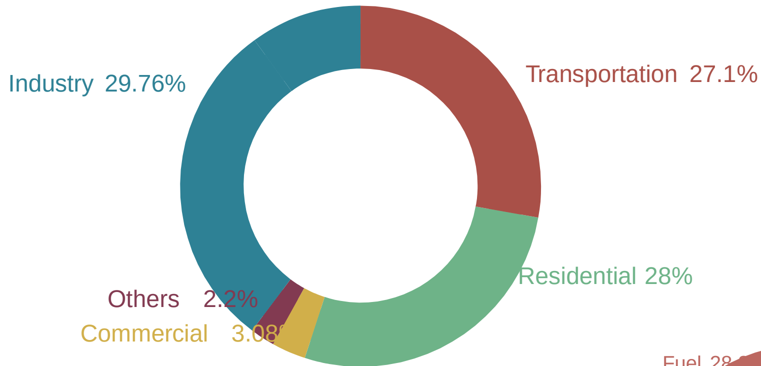
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC
**) Temporary data*

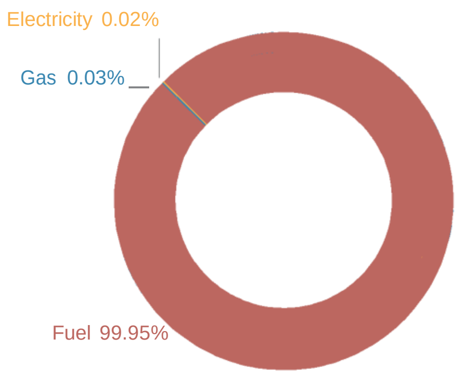
FINAL ENERGY CONSUMPTION



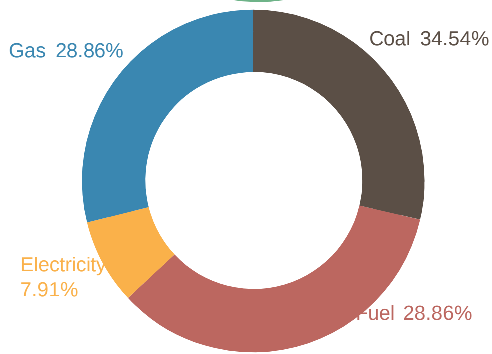
Source : BPS - Statistics Indonesia, reprocessed by NEC

SHARES OF ENERGY UTILIZATION BY SECTOR

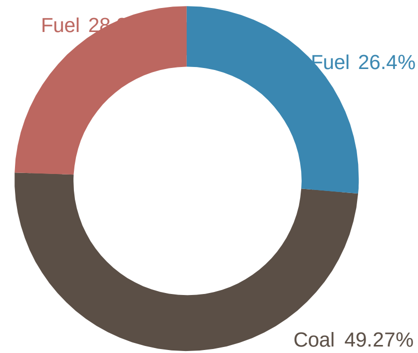




TRANSPORTATION



INDUSTRY



POWER PLANT

Source : BPS - Statistics Indonesia, reprocessed by NEC

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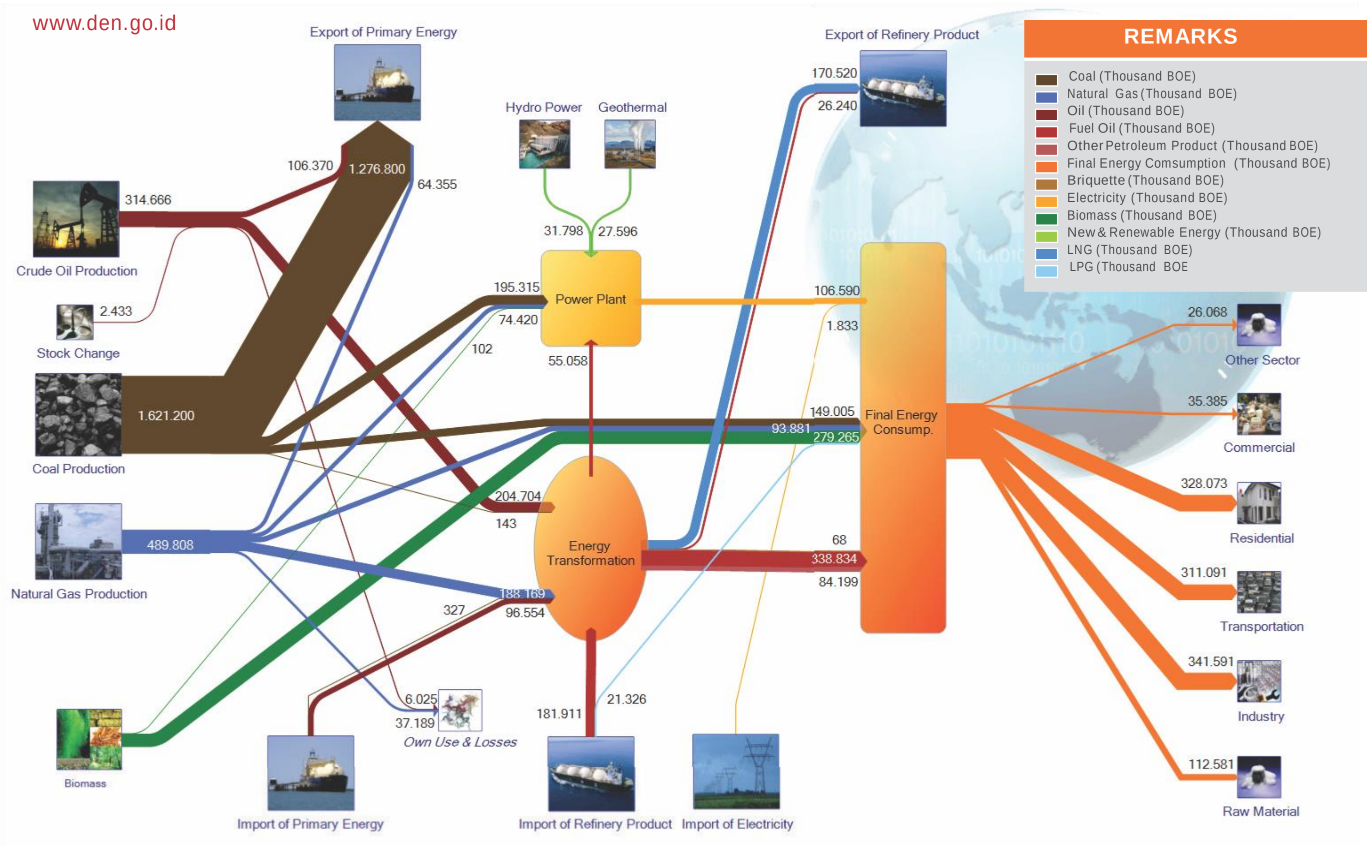
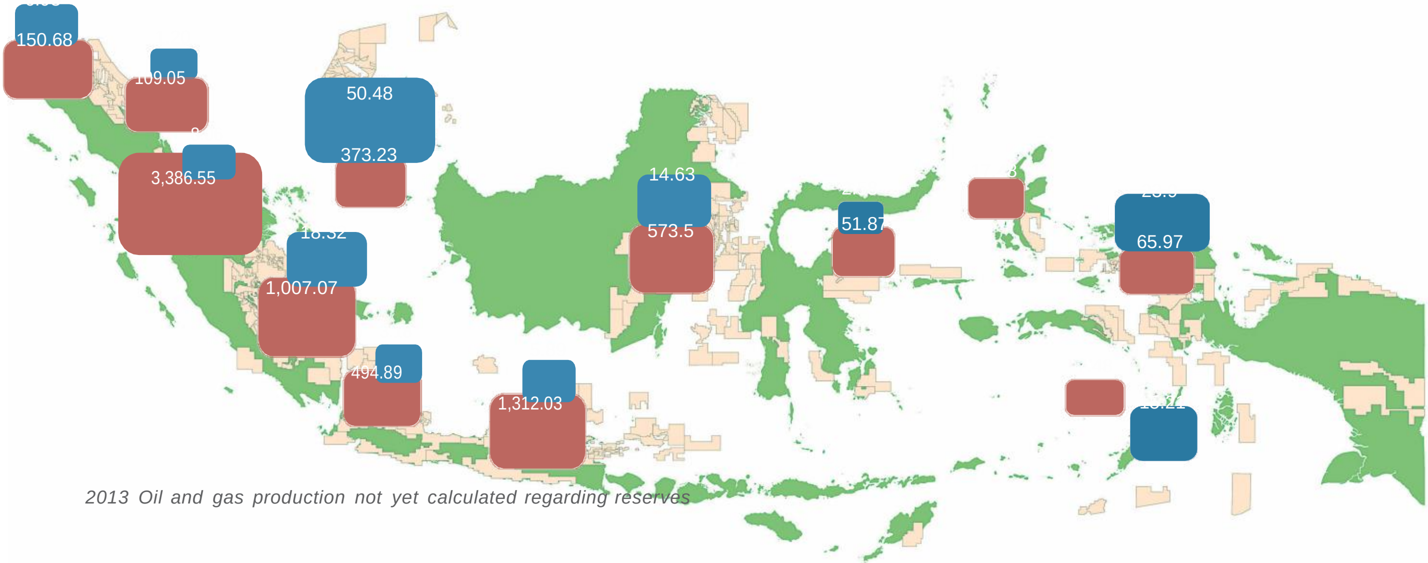
ENERGY FLOW, 2013*)



Source: National Energy Council *) in the process of updating data

OIL AND GAS

OIL AND GAS RESERVES



2013 Oil and gas production not yet calculated regarding reserves

PROVEN (P1)	=	3,692.49
POTENTIAL (P2+P3)	=	3,857.31
TOTAL (3P)	=	7,549.81

Compared to 2012

PROVEN (P1)	=	101.54
POTENTIAL (P2+P3)	=	48.85
TOTAL (3P)	=	150.39

Compared to 2012

Source: BPS - Statistics Indonesia, reprocessed by NEC

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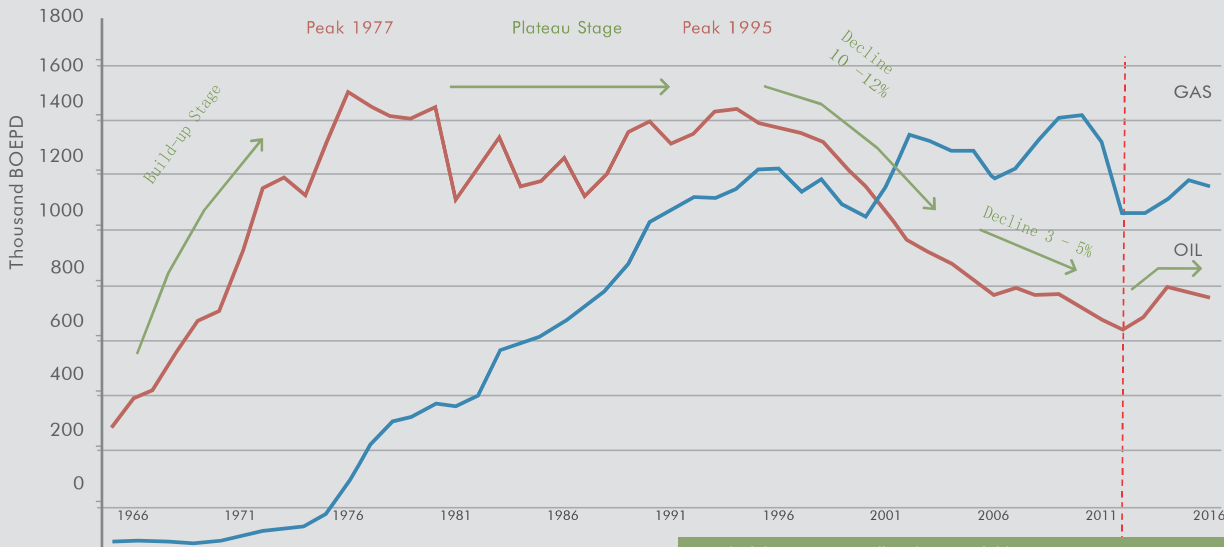
OIL AND GAS PRODUCTION PROFILE

OIL REMAINING RESERVE (MMSTB) | GAS REMAINING RESERVE (MMSTB)



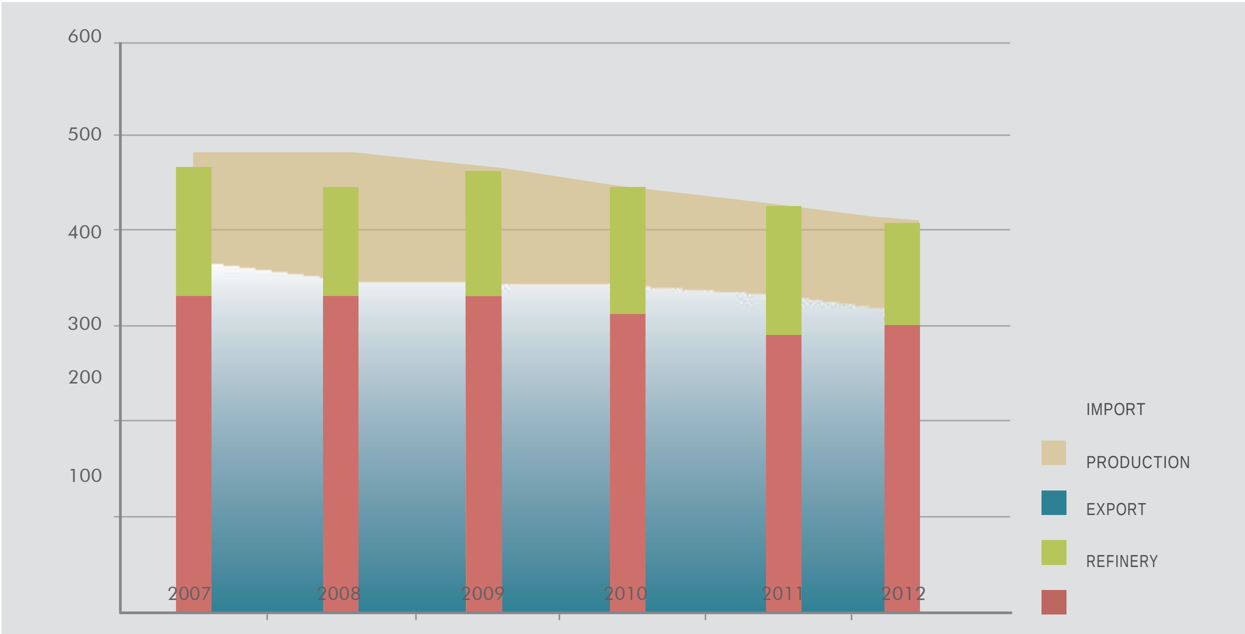
Source: SKK Migas by 2013, reprocessed by NEC

Trend of the upstream oil and gas activities

- Dominated by o shore locations
- The invention of reserves dominated by gas
- CBM start to developed
- Oil and gas exploration on the deep sea in the eastern region

CRUDE OIL SUPPLY

(Thousand Bbl)

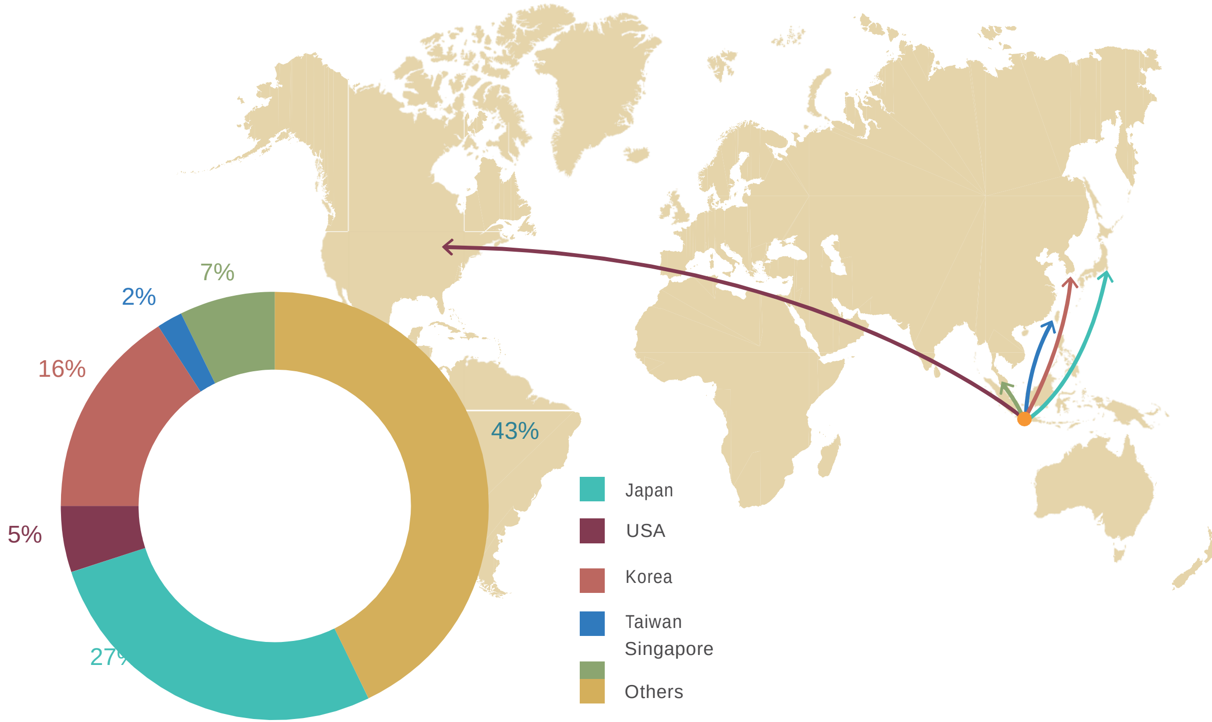


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Note : - Discrepancy between production, export, import and domestic demand due to the existing stock

- The role of imported crude in 2012 was amounted to 32 % from the total that goes to the oil re nery

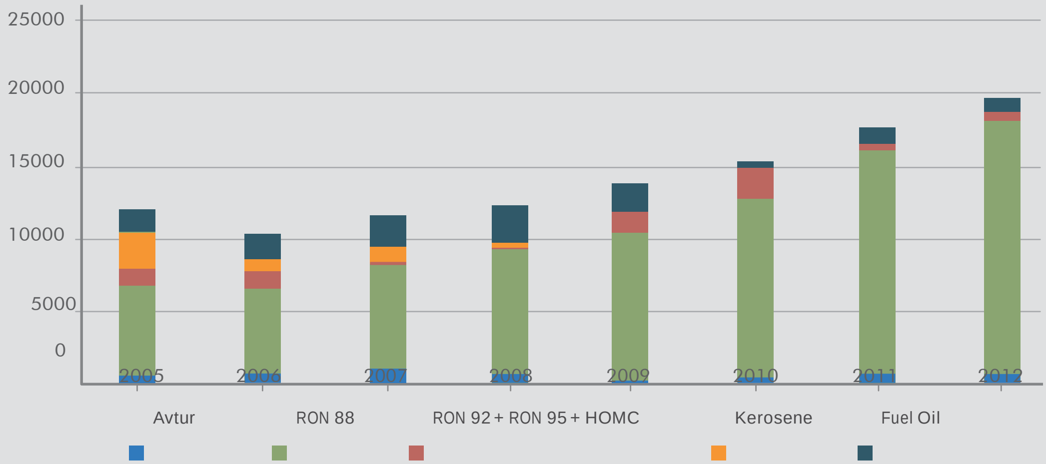
OIL AND GAS PRODUCTION PROFILE



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

FUEL IMPORTS

(Thousand KL)



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

Note :

- Share of fuel imports in 2012 amounted to 54.25% (179.98 MBOE) of the total domestic fuel oil consumption
- The value of imports of RON 88 (premium) increased along with the increase in domestic demand
- Kerosene imports declined and abolished in 2009 due to LPG conversion policy

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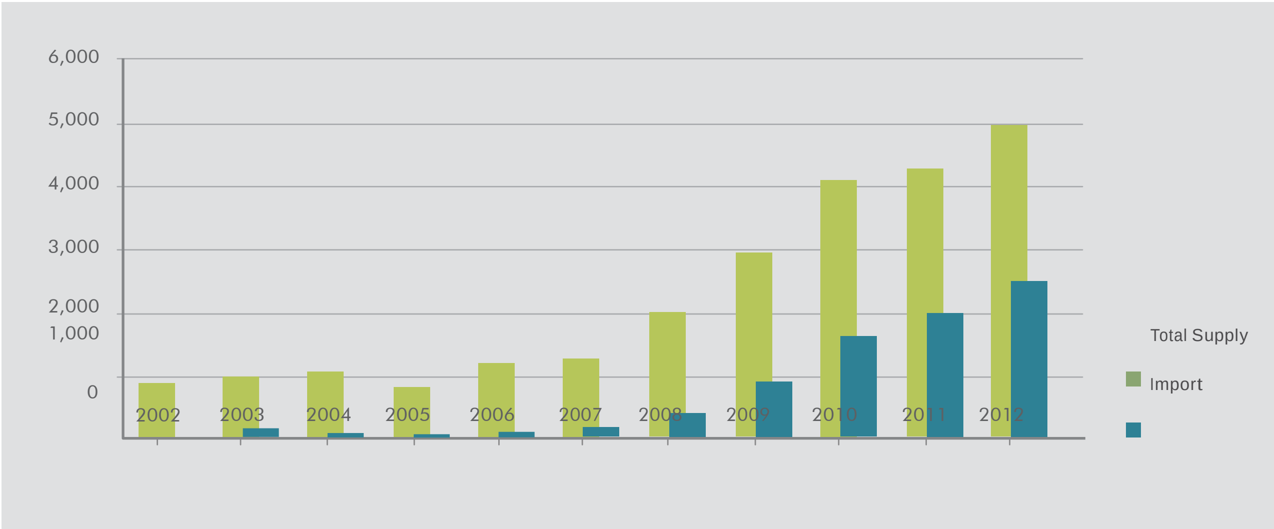
LPG IMPORTS

(Kilo Ton)



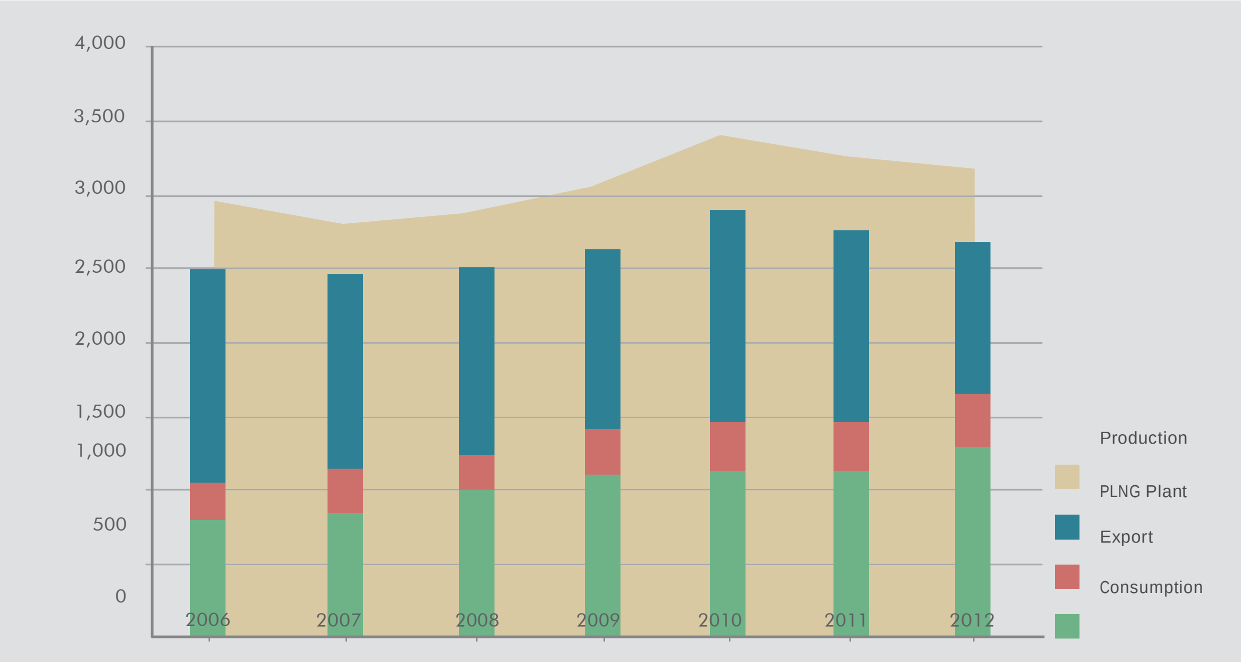
Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

Note :

- Share of LPG imports in 2012 amounted to 49.73% (21.3 MBOE) of the total LPG supply
- The value of LPG supply increased along with the increasing the demand in residential sector due to LPG conversion policy

GAS SUPPLY

(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

Note :

- Export means export of gas through pipeline
- Products of LNG re nery allocated for exports
- Discrepancy between the production and the total usage of gas due to gas reinjection, own use and are

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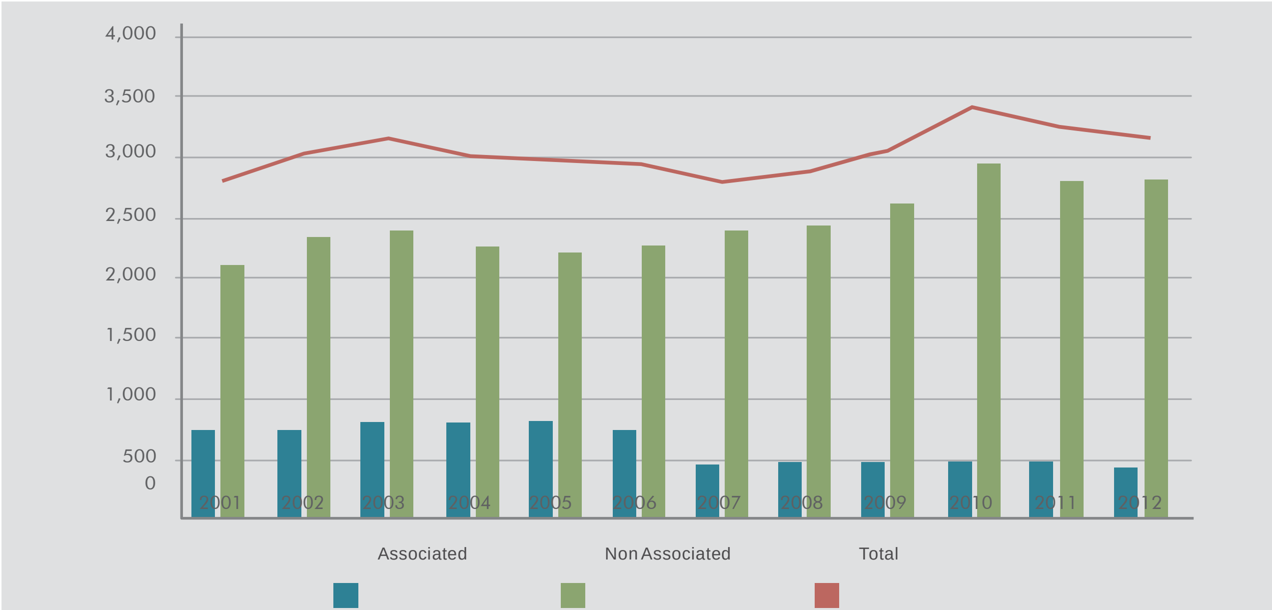
GAS PRODUCTION

(Thousand MMSCF)

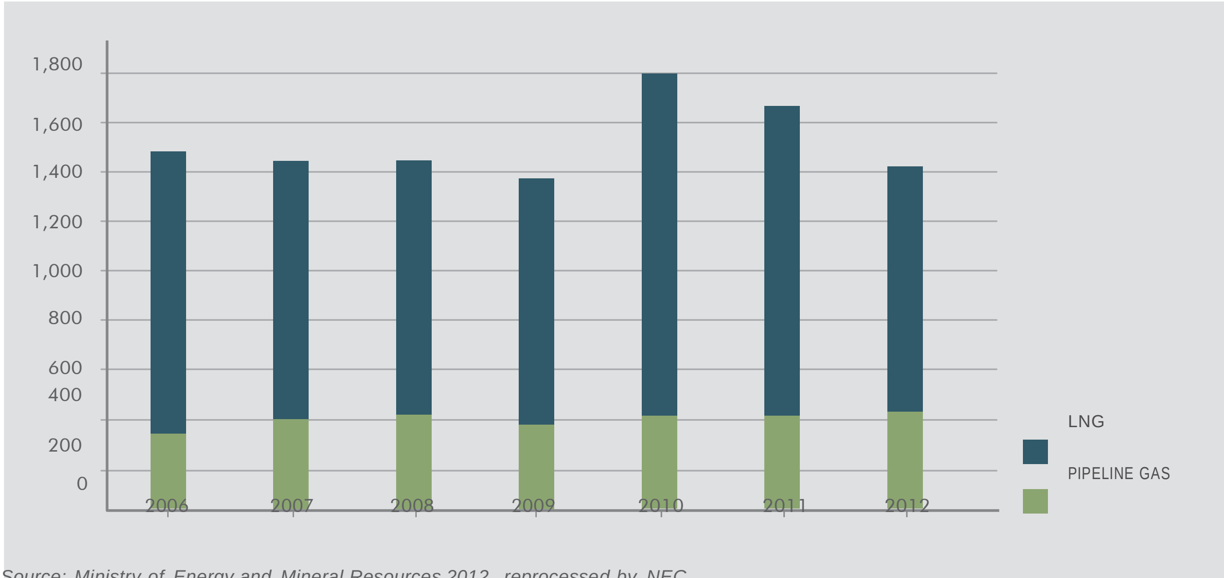


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Note :
Associated de ned as a natural gas found in association with oil, either dissolved in the oil or as a cap of free gas above the oil.

NATURAL GAS EXPORT

(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

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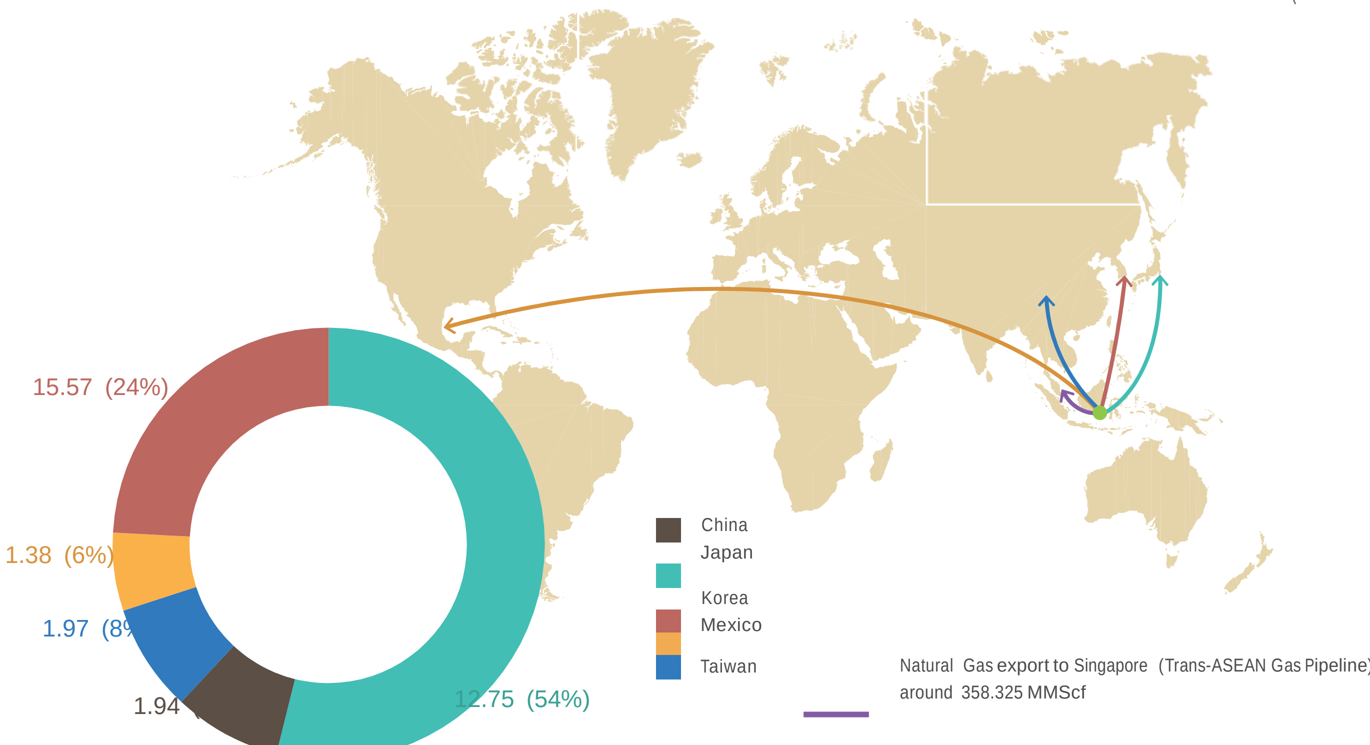
GAS AND LNG EXPORTS

(Million Ton)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

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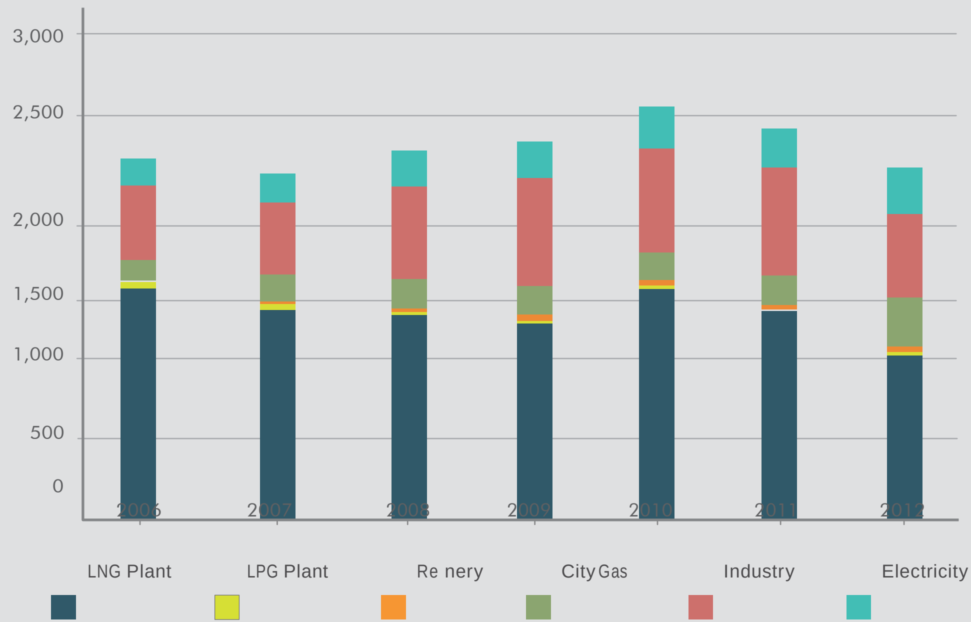
NATURAL GAS CONSUMPTION

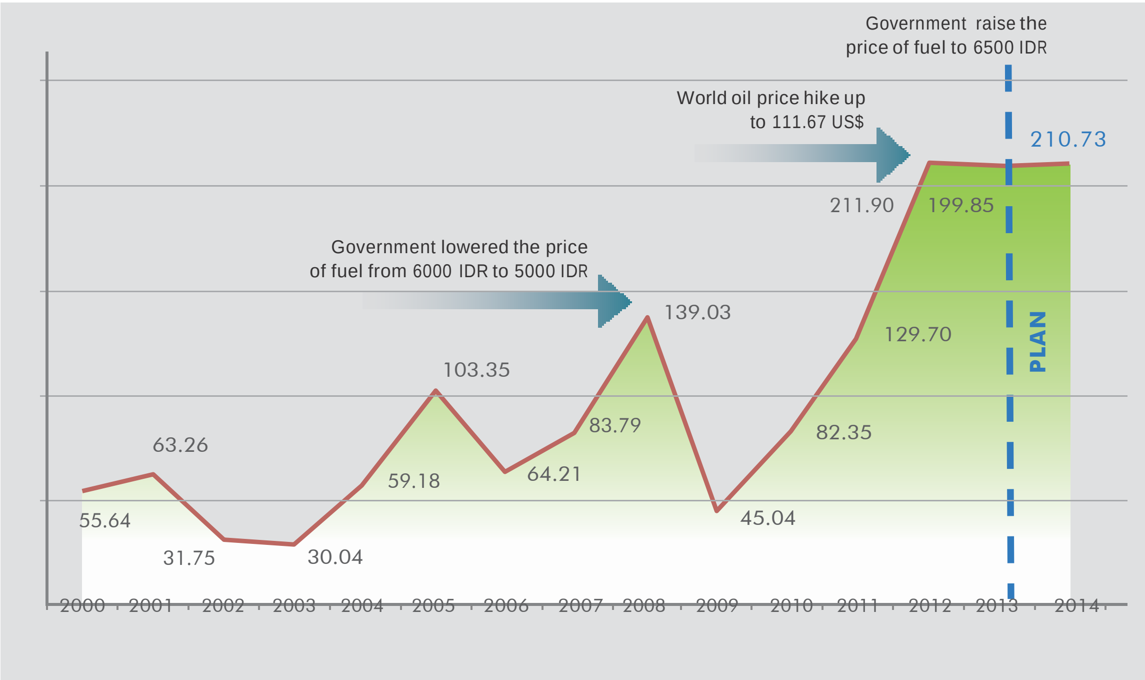
(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

FUEL OIL SUBSIDY

(Trillion IDR)

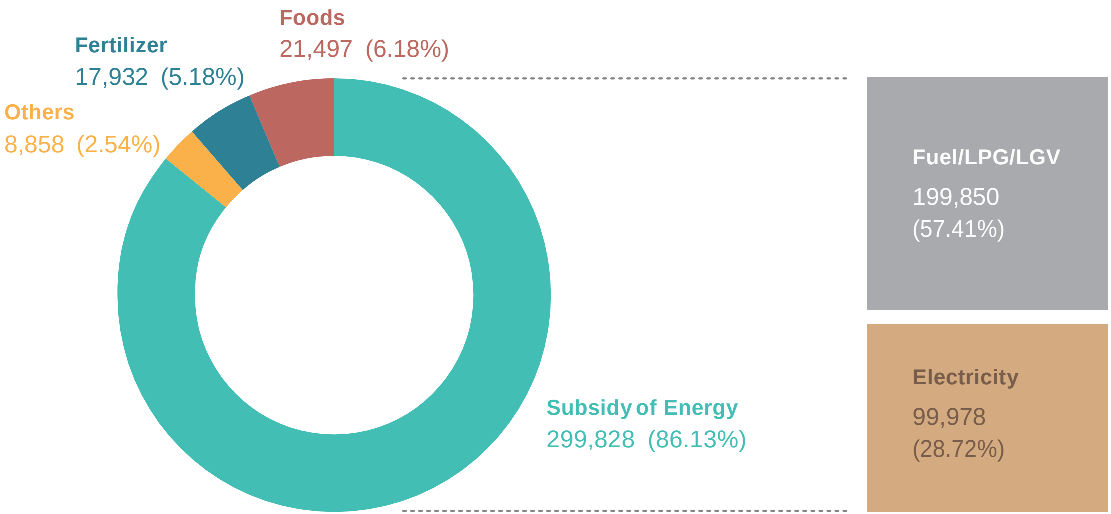


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Note: Rise in oil prices due to geopolitical crisis in the Middle East and Libya

SHARES OF TOTAL ENERGY SUBSIDY, 2013

(Trillion Rupiah)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

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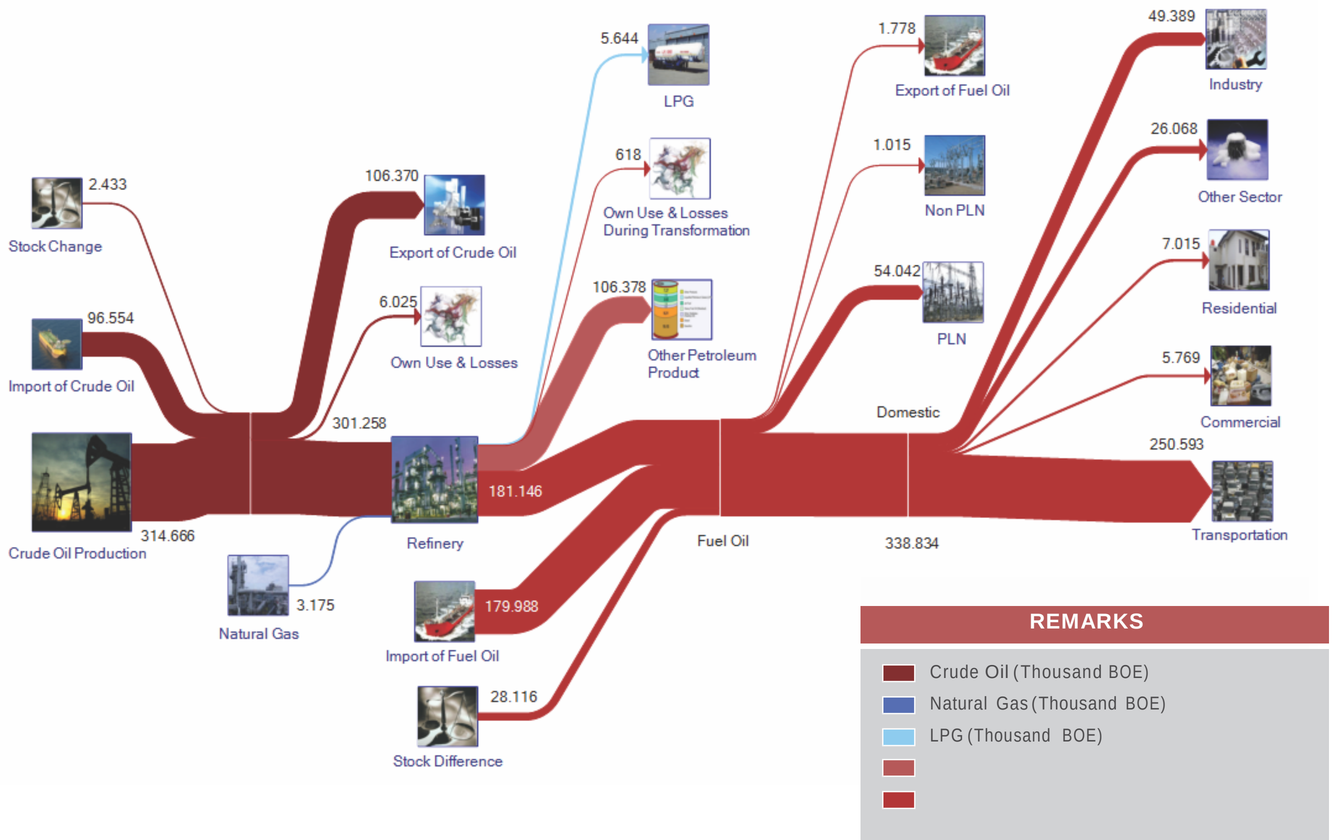
OIL FLOW, 2013*)



Other Petroleum Product (Thousand BOE)
Fuel Oil (Thousand BOE)

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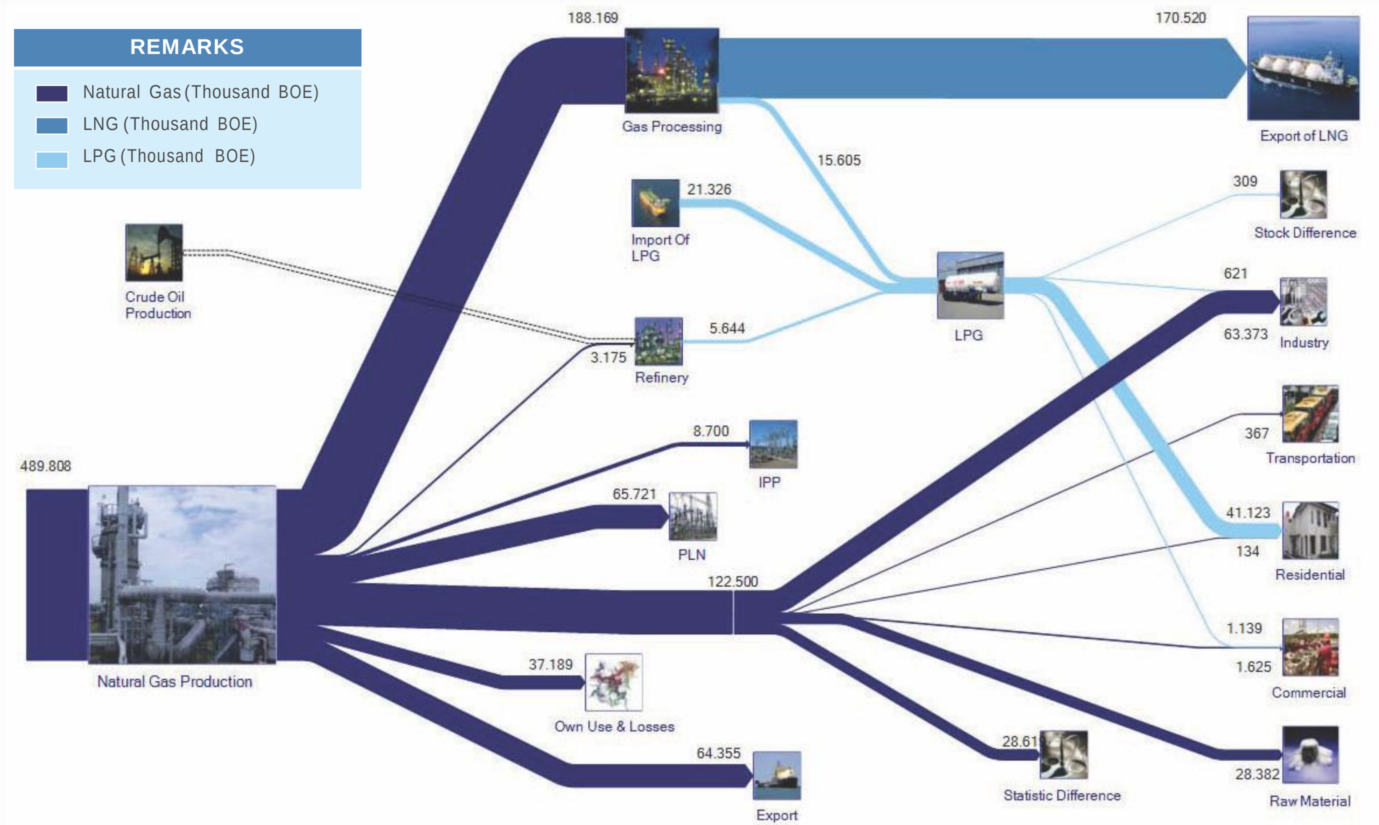
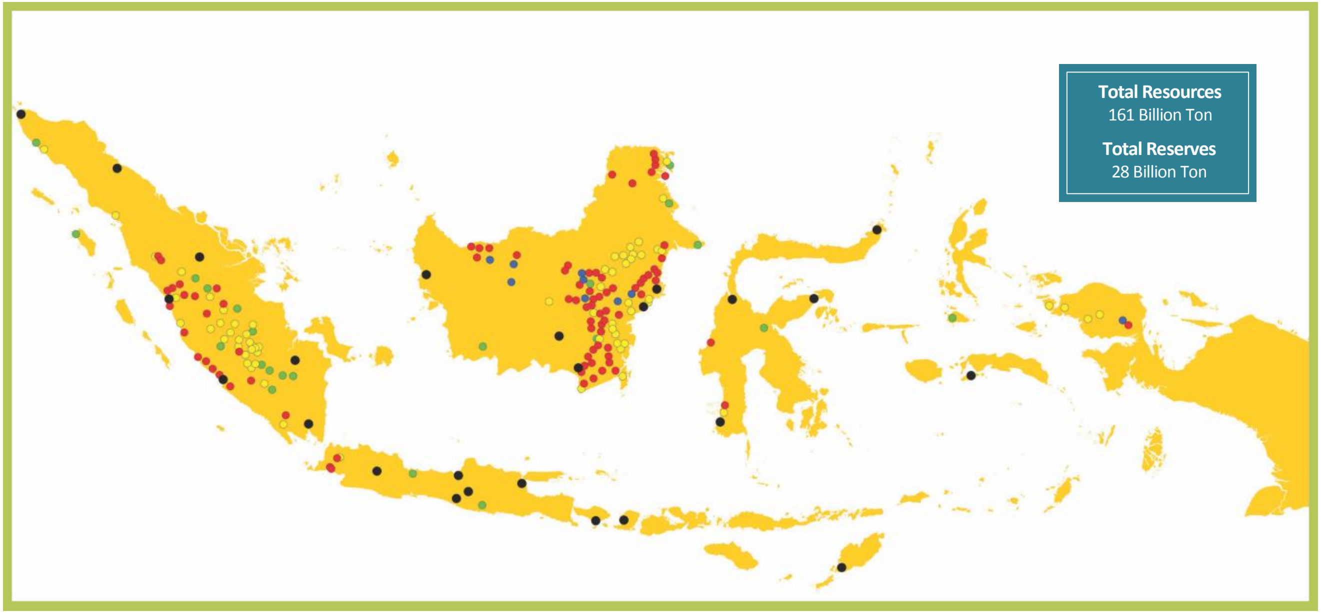
GAS FLOW, 2013*)



Source: *National Energy Council* *) *in the process of updating data*

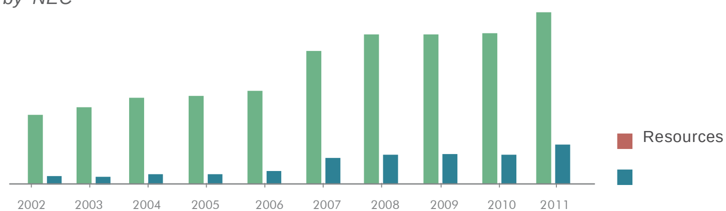
COAL

COAL RESOURCES AND RESERVES



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

- Very High (> 7,100 cal/gr)
- High (6,100 - 7,100 cal/gr)
- Medium (5,100 - 6,100 cal/gr)
- Low (< 5,100 cal/gr)
- Reserves

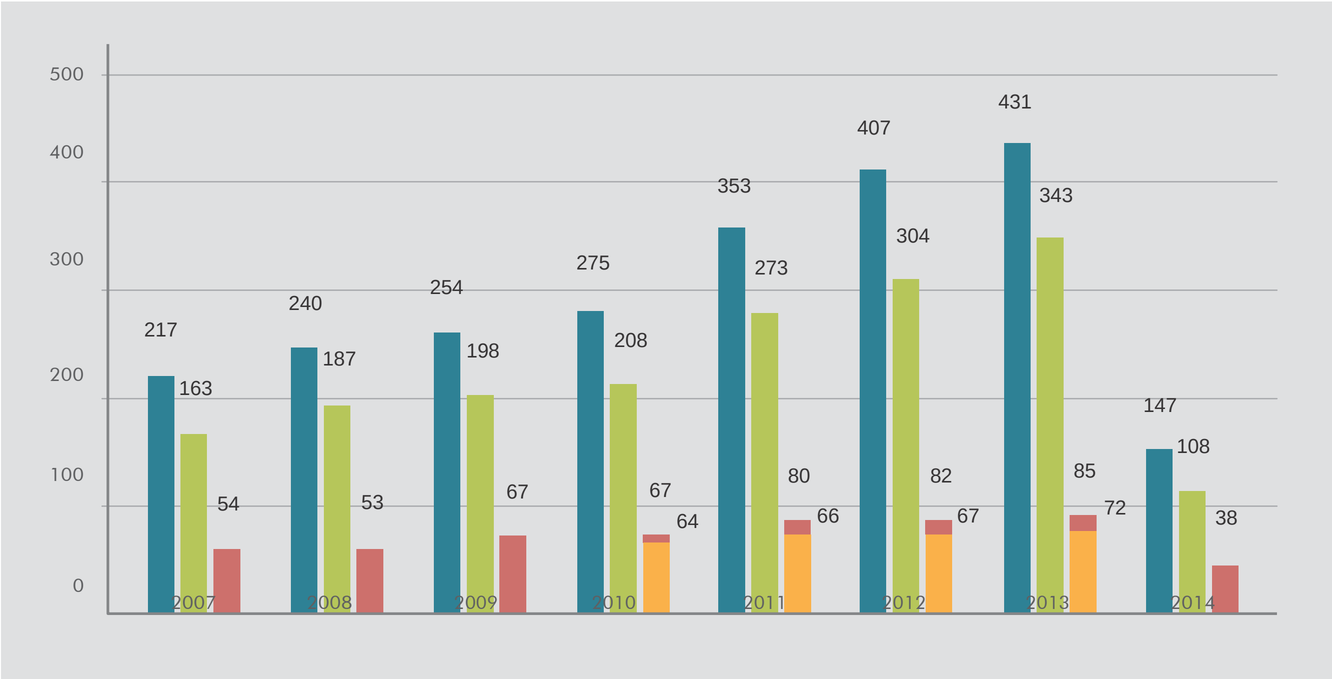


Reserves

2002 2003 2004 2005 2006 2007 2008 2009 2010 2011

COAL UTILIZATION

(Million Ton)



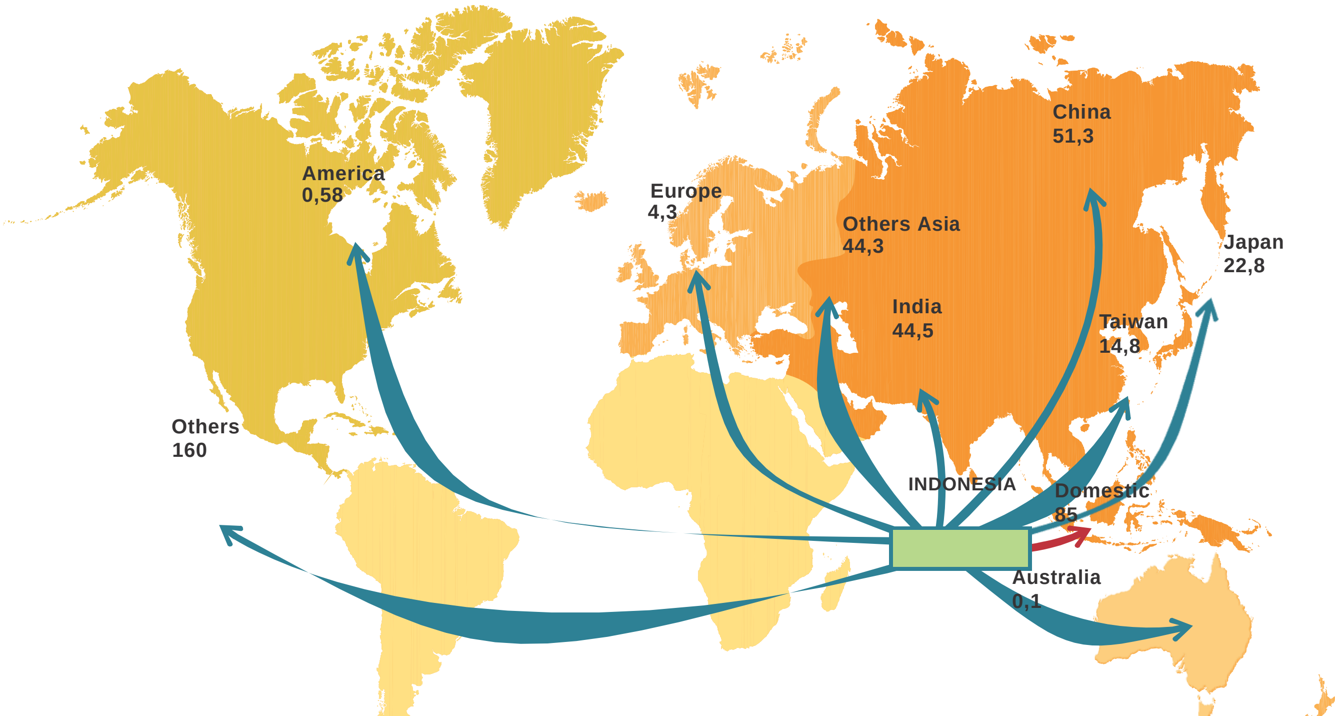
Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC
**) April 2014*

Production Export Domestic DMO

2002 2003 2004 2005 2006 2007 2008 2009 2010 2011

COAL EXPORT BY COUNTRY DESTINATION

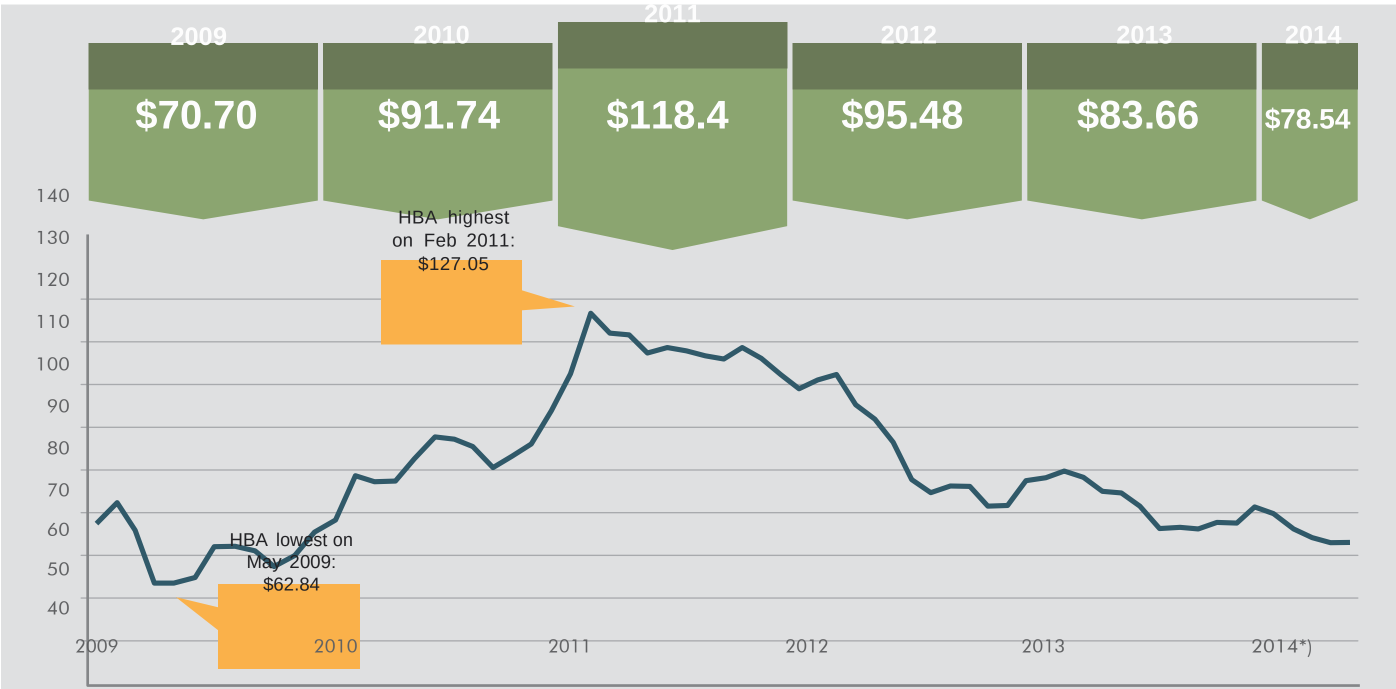
(Million Ton)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

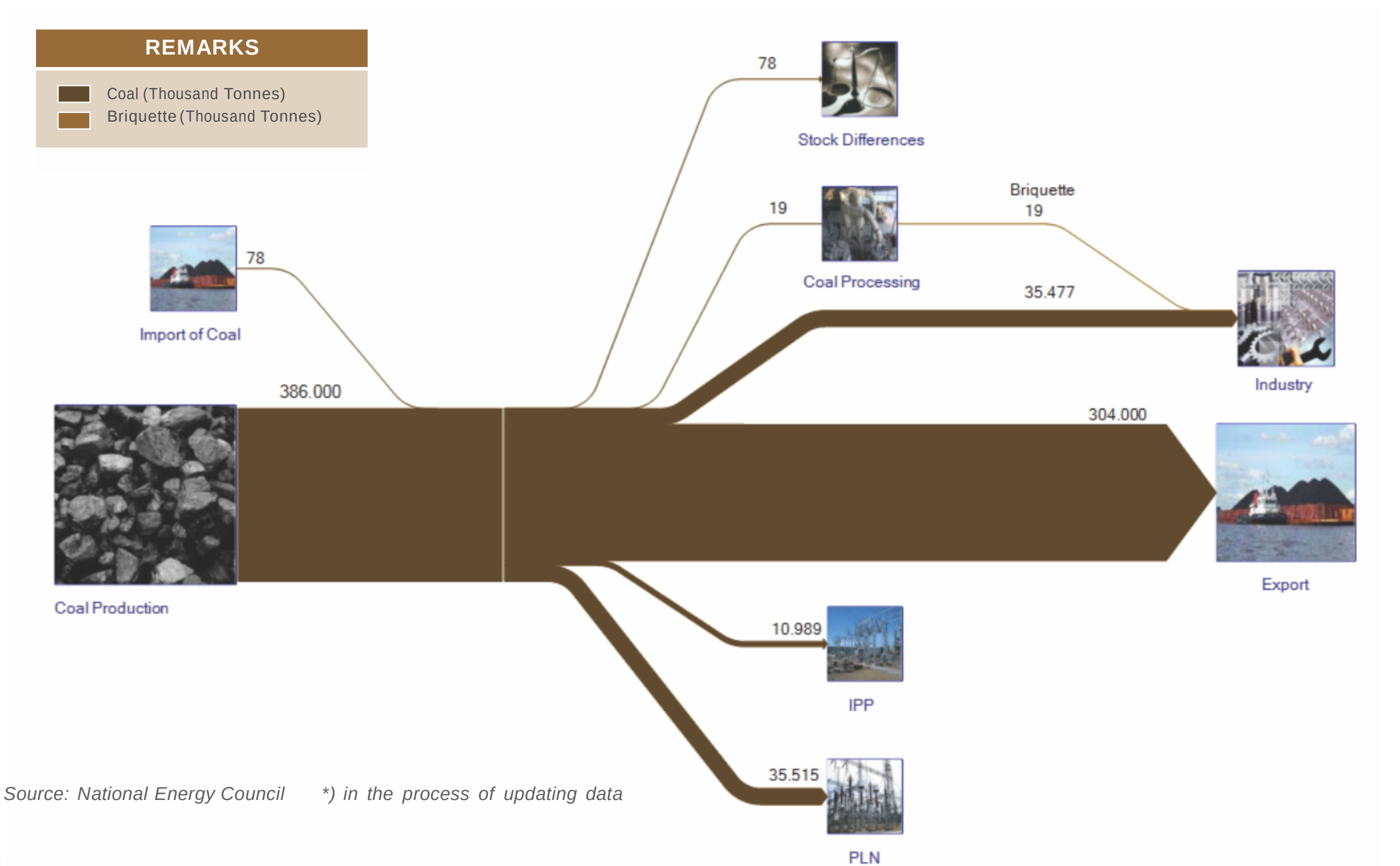
COAL REFERENCE PRICE (HBA)

(US Dollar)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC
**) Juni 2014*

COAL FLOW, 2013*)



Source: National Energy Council *) in the process of updating data

ELECTRICITY

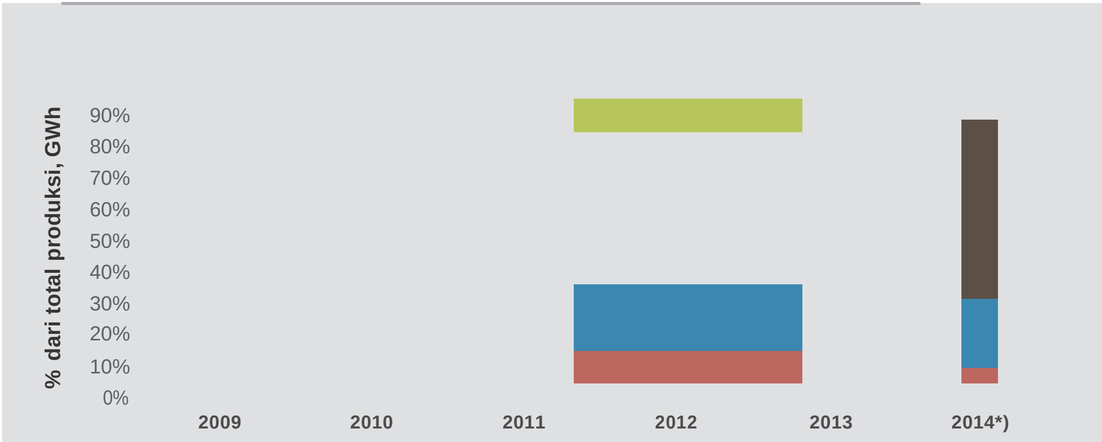
ELECTRICITY

DESCRIPTION	UNIT	2011	2012	2013
Demand Growth	%	10.1	8.4	8.6
Electri cation Ratio	%	72.9	76.6	80.5
Village Electri cation Ratio	%	96	96.7	97.8
Total Installed Capacity	MW	39,885	44,124	48,161
a. PLN	MW	30,529	32,108	35,564
b. IPP	MW	7,653	10,287	10,718
c. *Private Power Utilities* (PPU)	MW	1,704	1,729	1,729
PLN Electricity Production and Purchase of Electricity	GWh	175,213	193,663	207,409
Rural Electricity				
a. Sub Station Distribution	MVA	368	249	216.8
b. Distribution Network	KMs	17,570.7	11,311.5	9,244.3
c. Cheap and E cient Electricity	RTS*)	-	60,702	95,227

Source : Ministry of Energy and Mineral Resources & National Electric Company (PLN)
**) RTS = Target Household (<450 Watt)*

POWER PLANT ENERGY MIX

(%)



	2009	2010	2011	2012	2013	2014*)
Others RE	0%	0%	0.1%	0.1%	0.2%	0.5%
Geothermal	3%	3%	5.1%	4.9%	4.4%	4.4%
Hydro	8%	12%	6.8%	6.4%	7.7%	6.1%
Coal	39%	38%	44.1%	50.3%	51.6%	57.2%
Gas	25%	25%	21%	23.4%	23.6%	22.0%
Fuel Oil	25%	22%	23%	15%	12.5%	9.7%

Source : Ministry of Energy and Mineral Resources, reprocessed by NEC
**) Target 2014*

POWER PLANTS DEVELOPMENT PLANNING

Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Hydropower Plant
Coal Gasi cation Power Plant
Minihydro Power Plant
Steam Power Plant
Gas Power Plant
Steam Gas Power Plant
Geothermal Power Plant
Micro Hydro Power Plant

ELECTRICITY TRANSMISSION NETWORK

Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Existing Transmission Line

Transmission Line Plan

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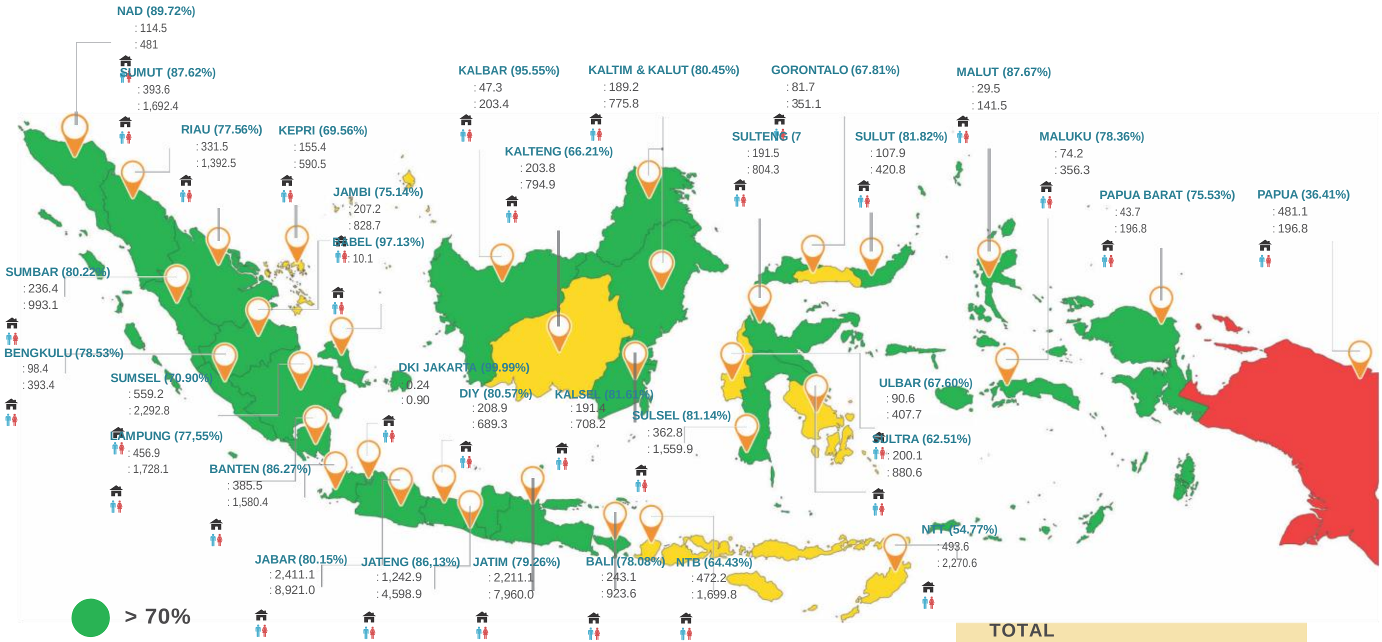
ELECTRIFICATION RATIO



> 70%

> 50- 70%

< 50%

Non Electri cation (Thousand)

Population Not Electric (Thousand)

TOTAL

RE : 80.51%

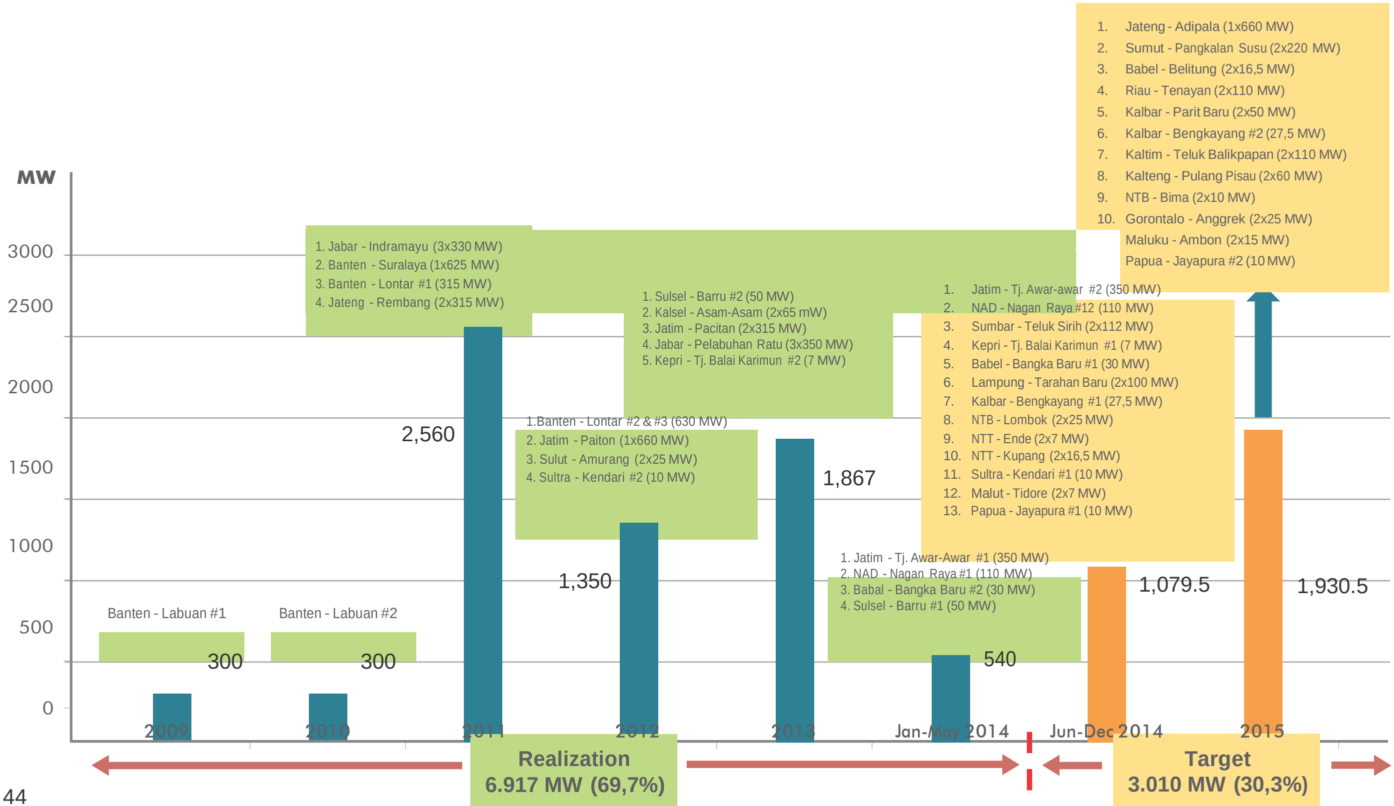
: 12,530,301 Household

: 48,868,174 People's

Source: Ministry of Energy and Mineral Resources and BPS, reprocessed by NEC

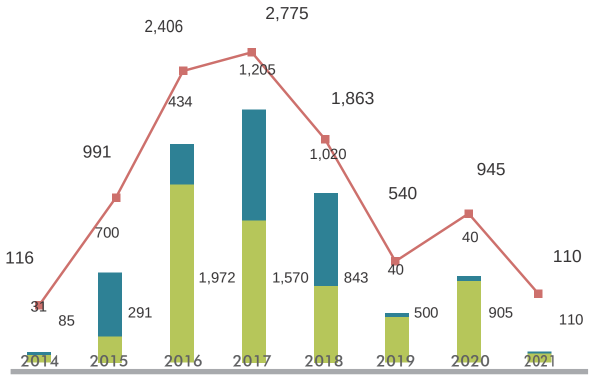
PROGRESS OF FTP-I AND FTP-II

REALIZATION AND COD TARGET OF POWER PLANT
10.000 MW FAST TRACK PROGRAM PHASE I

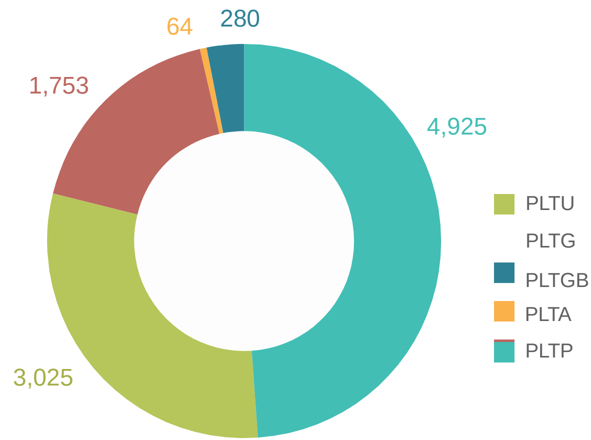


10,000 MW FAST TRACK PROGRAM PHASE II (FTP II)

COMMERCIAL OPERATION PLAN (MW)

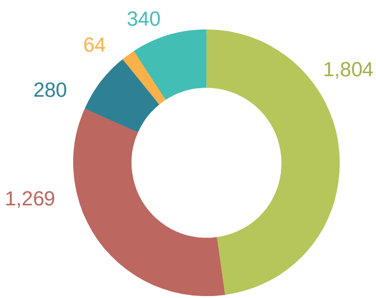


BY TYPE OF POWERPLANT (MW)

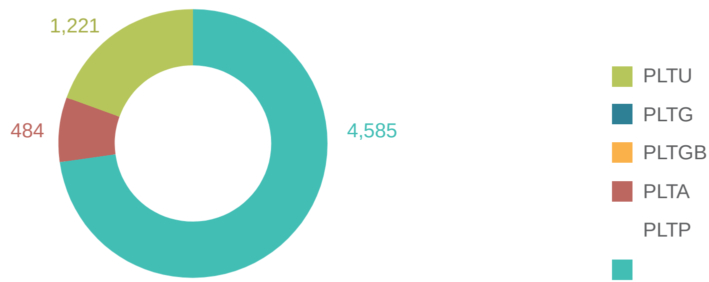


KOMPOSISI PER JENIS PEMBANGKIT

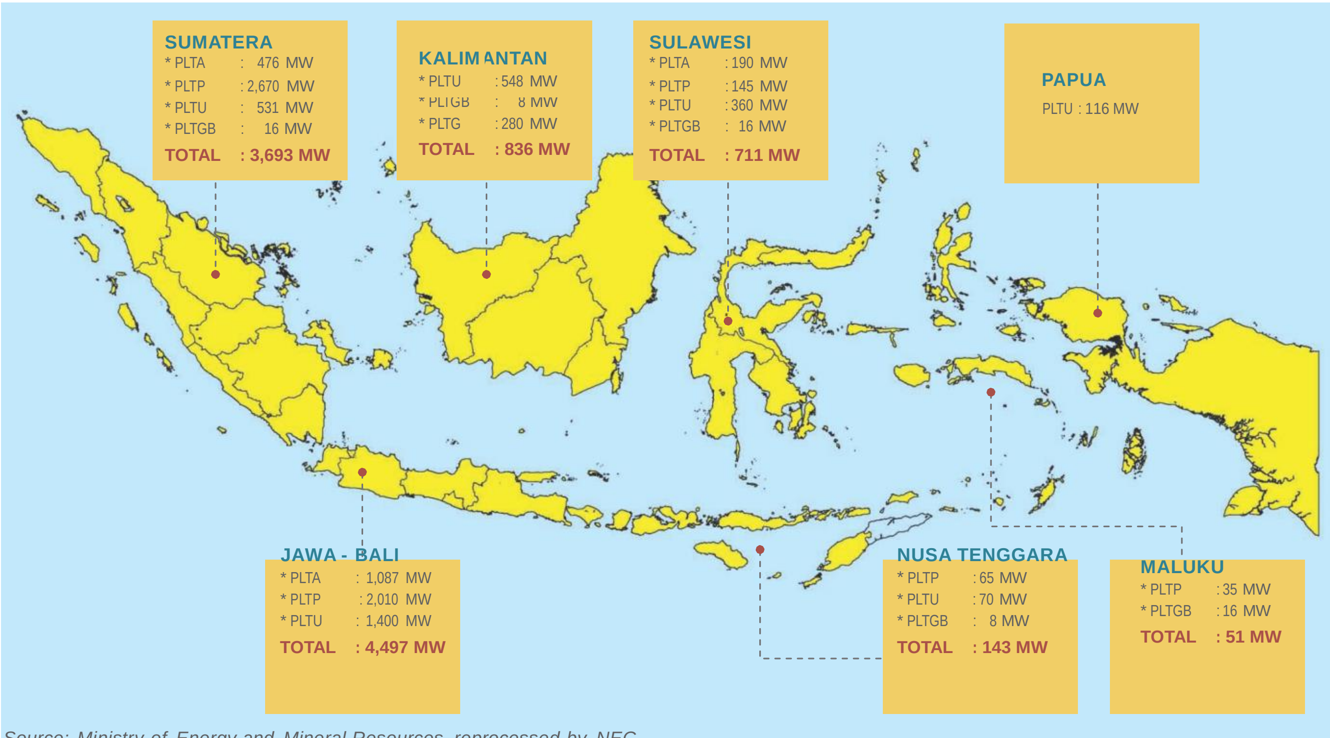
PLN (26 PROJECTS) : 3.757 MW (37%)



IPP (72 PROJECTS) : 6.290 MW (63%)



DISTRIBUTION OF 10,000 MW PROGRAM PHASE II



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC
**) EMR Ministerial Regulation Number 01 year 2012*

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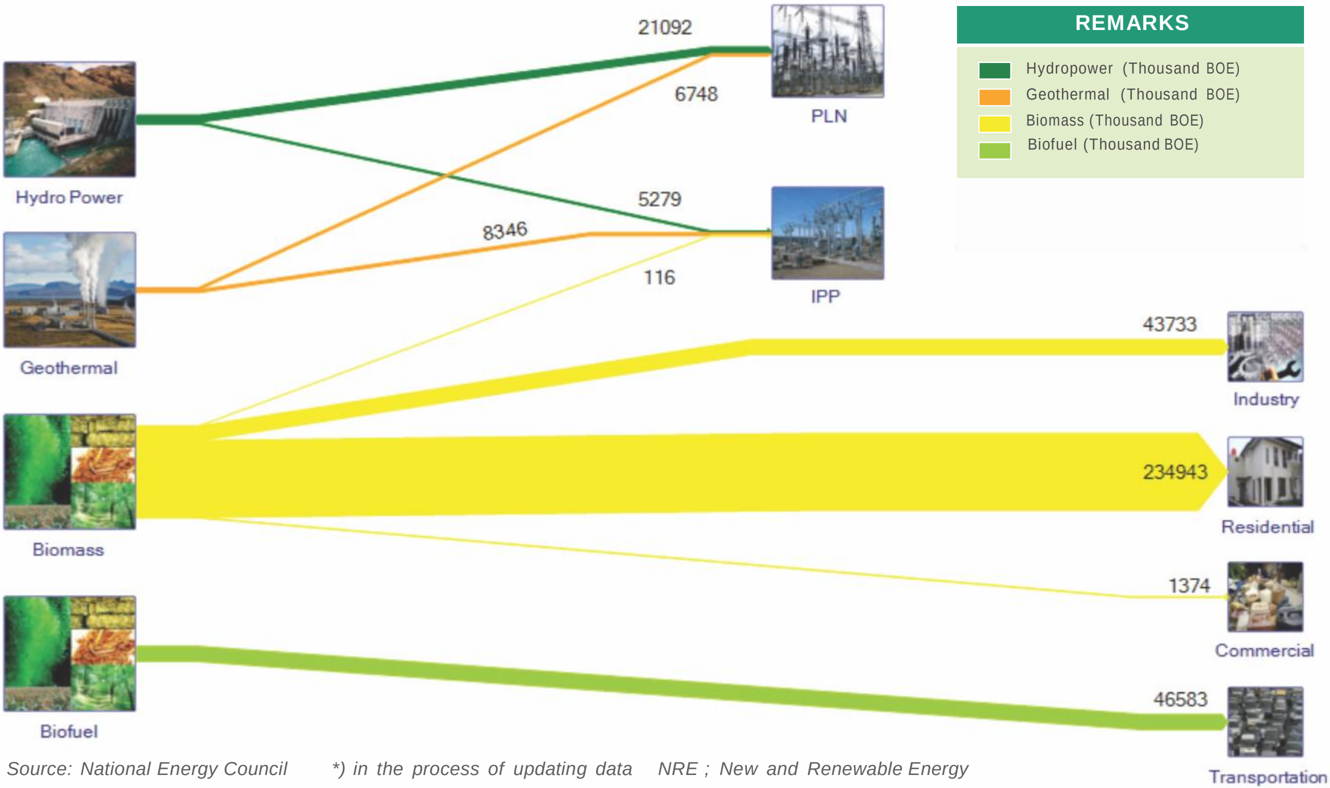
NEW AND RENEWABLE ENERGY

N R E FLOW, 2013*)



Source: National Energy Council **) in the process of updating data* *NRE ; New and Renewable Energy*

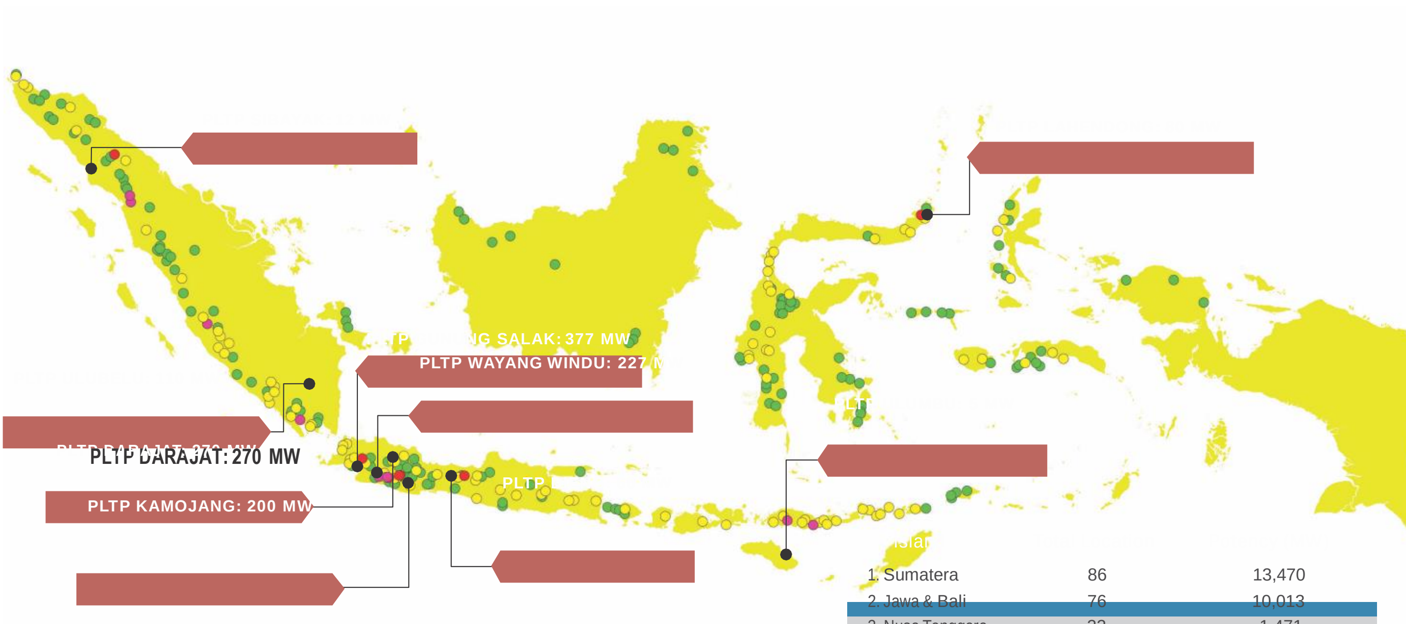
NEW AND RENEWABLE ENERGY RESOURCES

NO.	TYPE	RESOURCE	INSTALLED CAPACITY (MW)	RATIO (%)
1	2	3	4	5 = 4/3
1	Hydro (MW)	75,000MW	6,848.46 MW	9.13%
2	Geothermal (MW)	29,164 MW	1,341 MW	4.6 %
3	Biomass (MW)	49,810 MW	1,644.1 MW	3.3%
4	Solar Energy	4.80 kWh/m^2/day	22.45MW	-
5	Wind Energy	3 –6 m/s	1.87 MW	-
6	Ocean	49 GW 2)	0.01 MW3)	0%
7	Uranium(MW)	3,000 MW	30 MW1)	0%

Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Note:
1) Research scale, non-energy
2) Source: National Energy Council
3) Research Scale: BPPT

GEOTHERMAL POTENTIAL



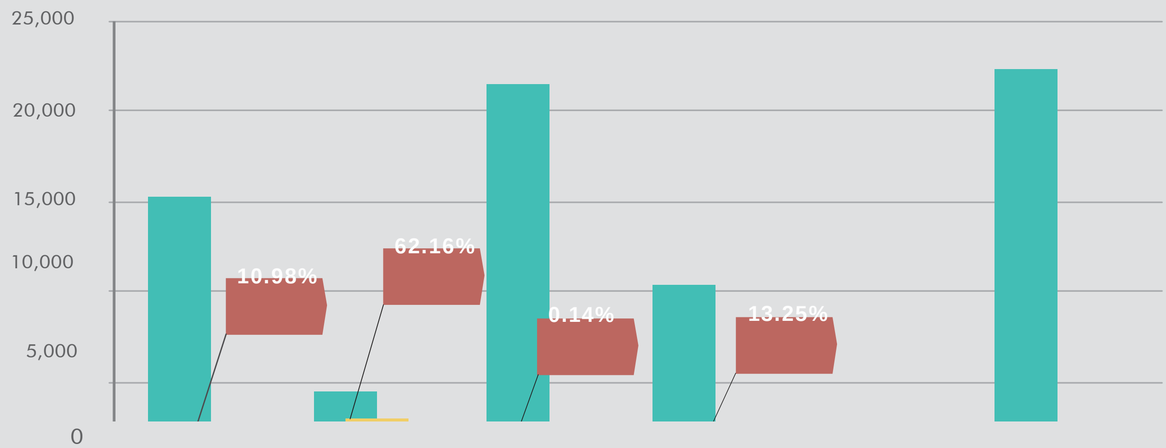
1. Sumatera	86	13,470
2. Jawa & Bali	76	10,013
3. Nusa Tenggara	22	1,471
4. Kalimantan	12	145
5. Sulawesi	56	2,939
6. Maluku & Papua	33	1,126
TOTAL	285	29,164

Source: Geological Agency, Ministry of Energy and Mineral Resources (2012)

- Preliminary Survey
- Ready to be developed
- Detailed Survey
- Installed

HYDRO POTENTIAL AND INSTALLED CAPACITY

(MW)



	Sumatera	Java	Kalimantan	Sulawesi	Bali, Nusa Tenggara	Maluku, Papua
Potential	15,600	4,200	21,600	10,200	620	22,780
Installed Capacity	1,772.60	2,652.40	31.34	1,460.41	4.65	8.59

Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

- *Total hydropower potential 75 GW, installed capacity 5,940MW (7.9% of potential)*
- *Installed capacity data included minihydro and microhydro powerplant.*

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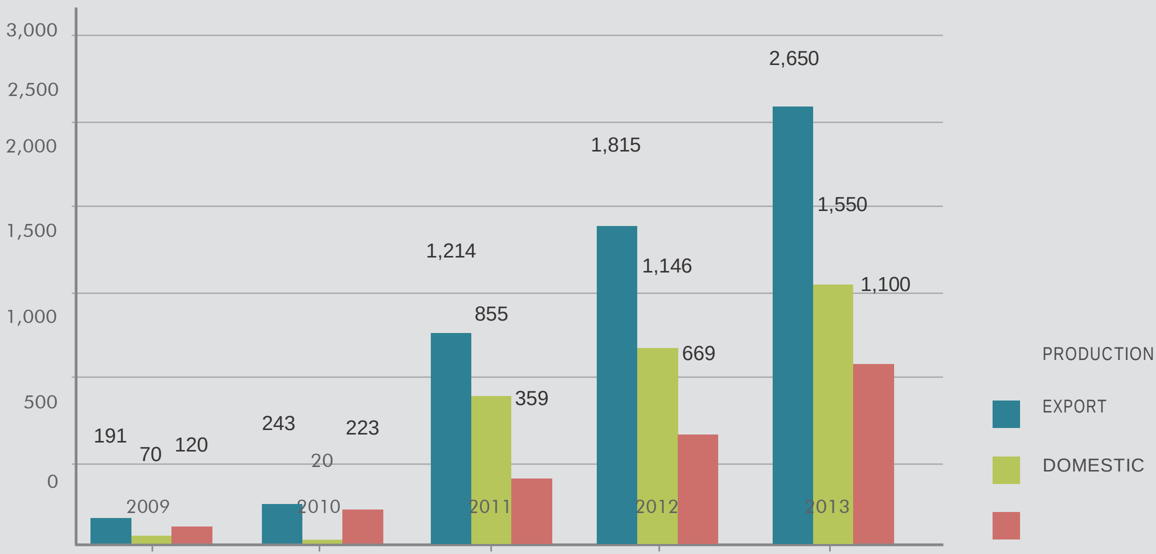
BIO-FUEL PRODUCTION

(Thousand KL)

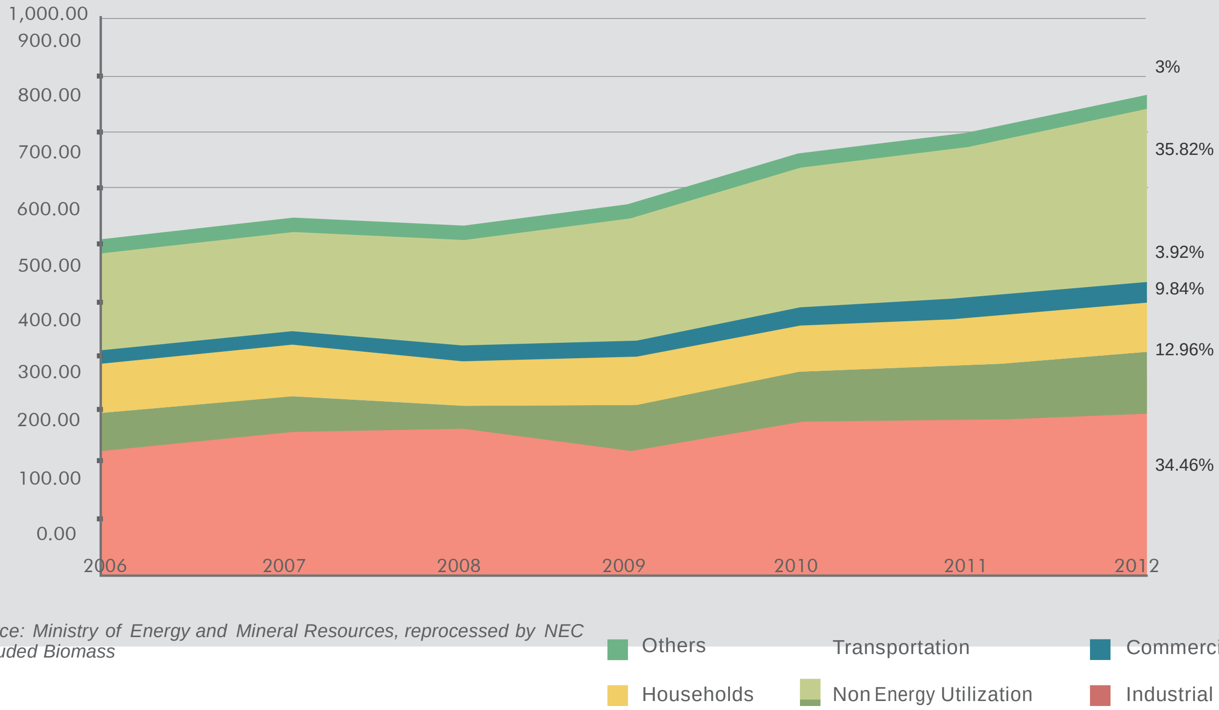


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

ENERGY CONSUMPTION BY SECTOR

BIO-FUEL PRODUCTION

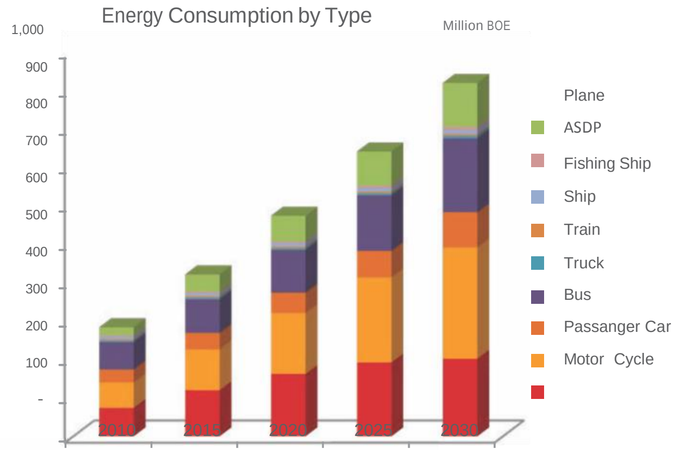
(MBOE)

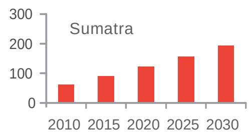


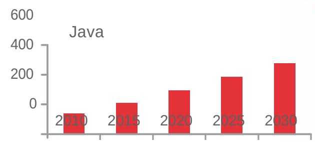
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC
Excluded Biomass

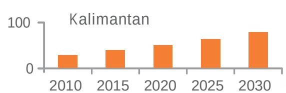
TRANSPORTATION

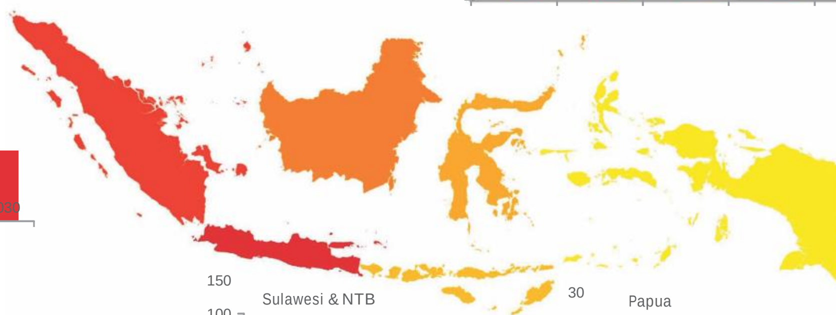
TRANSPORTATION					
Capacity (people)	1,500	40	500	2	1,500
Consumption (Liter/Km)	3	0.5	40	0.08	10
Consumption (Liter/KM/People)	0.002	0.0125	0.05	0.04	0.006

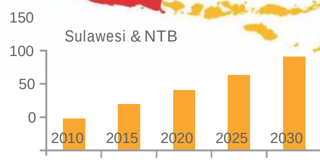


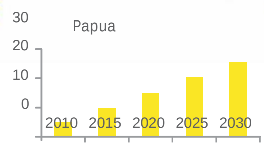






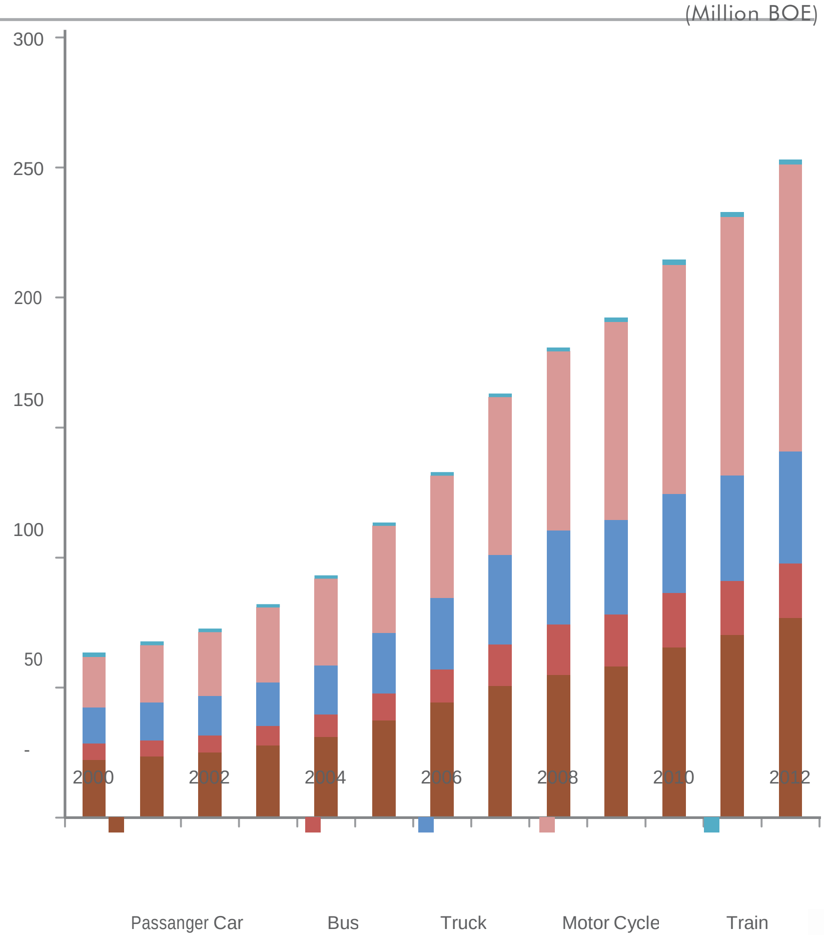






Source: Ministry of Transportation, reprocessed by NEC

ENERGY CONSUMPTION VEHICLES BY TYPES



Passenger Car
Subsidized fuel consumption for passenger cars are very huge, the total fuel subsidized by the Government almost 30% consumed by passenger cars, and 44% by motorcycles.

Bus
Bus consumed 8% from total fuel subsidized by government.

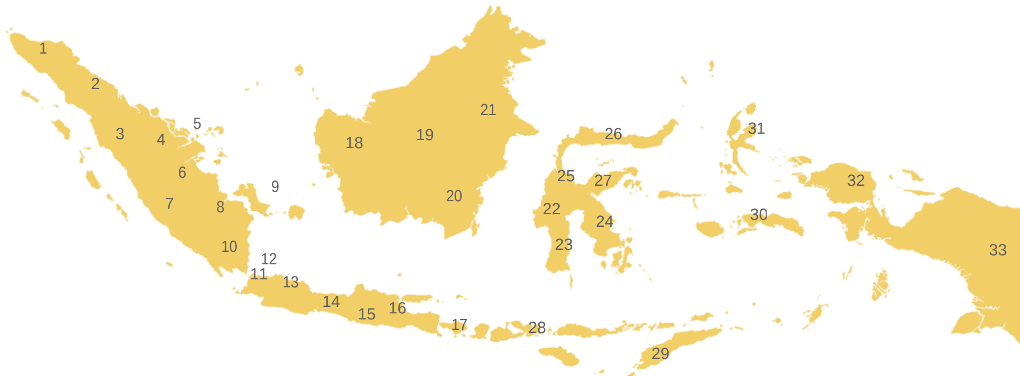
Truck
Truck consumed 17% from total fuel subsidized by government.

Motor Cycle
Motor cycle growth rate is very high (15.3% per year), exceed the growth rate of total vehicles by 14% per year.

Train
Train consumed only 1% from the total fuel. To date, trains service are only available in Java and Sumatera islands.

Source: Ministry of Transportation, reprocessed by NEC

ENERGY CONSUMPTION BY INDUSTRY



1. NAD	2. Sumatera Utara	3. Sumatera Barat	4. Riau	5. Kepulauan Riau	6. Jambi	7. Bengkulu	8. Sumatera Selatan	9. Bangka Belitung	10. Lampung	11. Banten	12. DKI Jakarta	13. Jawa Barat
17588	132431	29421	175787	231052	177926	6835	61765	416092	83057	417570	163327	819689
5	174913	1010841	286815	5460	68770	0	206000	62480	212653	646144	41164	1924763
101	14937	2	13503	5336	48	0	10836	7	202	53399	15551	82267
16118	455356	304363	103517	1035949	121565	2771	119278	6604	81474	2224937	1959680	7852385

14. Jawa Tengah	15. DI Yogyakarta	16. Jawa Timur	17. Bali	18. Kalimantan Barat	19. Kalimantan Tengah	20. Kalimantan Selatan	21. Kalimantan Timur	22. Sulawesi Barat	23. Sulawesi Selatan	24. Sulawesi Tenggara	25. Sulawesi Tengah	26. Sulawesi Utara
131883	25705	524510	9998	36848	72391	66992	70442	4071	39125	12678	5056	39125
530599	131604	663113	9371	17357	53	124757	42805	3	1254647	171477	1871	1254647
1347	11	153474	17	5	30	1	12	-	82	3	20	82
1775626	128090	3095922	21370	11718	3158	32071	90139	155	342012	879	5398	342012

27. Gorontalo	28. Nusa Tenggara Barat	29. Nusa Tenggara Timur	30. Maluku	31. Maluku Utara	32. Papua Barat	33. Papua
918	1118	39125	6075	16	64810	15285
226	41	1254647	-	-	599	39
-	17	82	3	-	7042	29
261	1892	342012	977	54	4995	1333

Total

Fuel Oil (KiloLiter)	: 3,874,768
Coal (Ton)	: 7,648,460
Gas (Thousand M^3)	: 358,288
Electricity (Thousand KwH)	: 1,984,178

Source: Ministry of Transportation, reprocessed by NEC

Passanger Car Bus Truck Motor Cycle Train

ENERGY DEMAND PROJECTION (2012-2017)

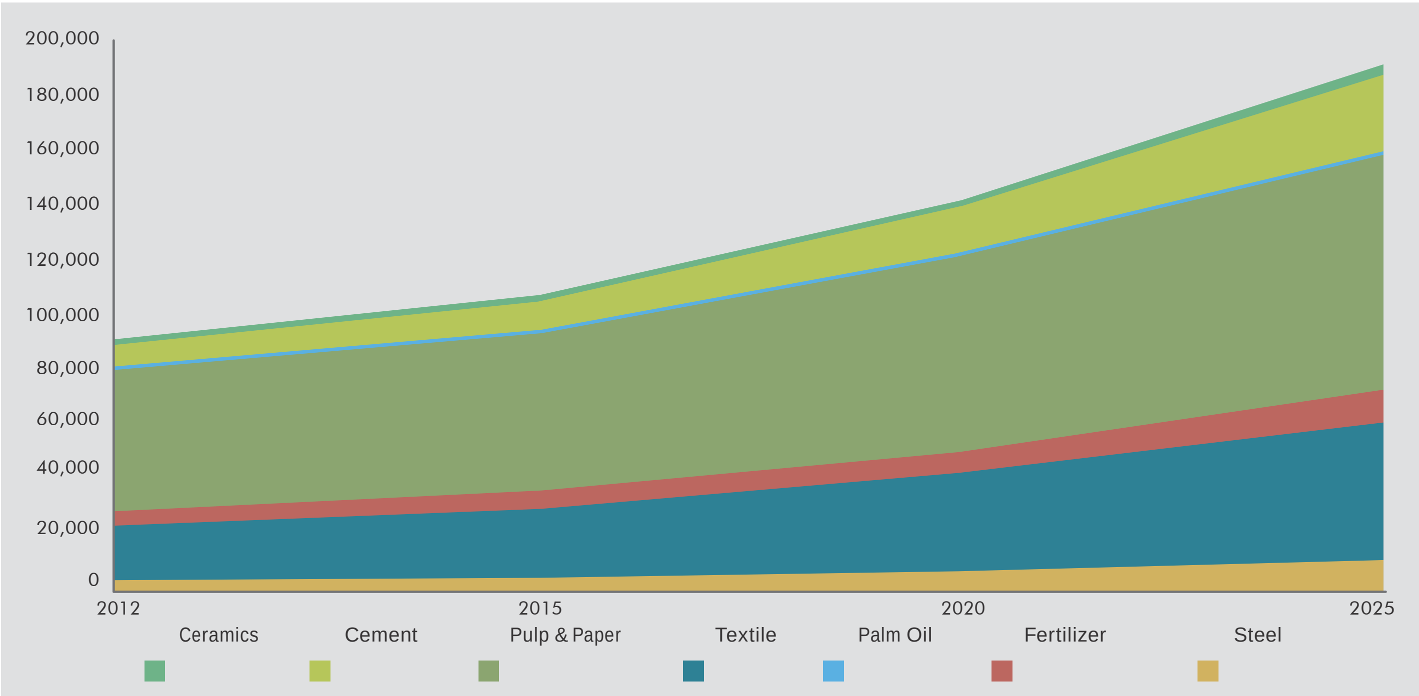
1. KS-POSCO Cilegon Steel Industry	(700 MW)
2. Subang Hi-Tech Industry	(600 MW)
3. Cilamaya Otomotif	(220 MW)
4. Majalengka Textile	(175,4 MW)
5. Kendal Textile	(159,12 MW)
6. Demak Alumina	(60 MW)
7. Kulonprogo Steel	(529,2 MW)
8. Boyolali Textile	(56,4 MW)
9. Tuban Chemical	(40 MW)
10. Jombang Footwear	(163,64 MW)
11. Gresik Petrochemical	(57 MW)
12. Lamongan Shipment	(190 MW)
13. Bintuni Oil & Gas	(400 MW)
14. Tangguh Oil & Gas	(430,4 MW)
15. Bitung Warehouse	(122 MW)
16. Palu Rattan	(300 MW)
17. Takalar Ferronickel	(1.000 MW)
18. Gowa Cocoa	(168,42 MW)
19. Jeneponto Salt	(170 MW)
20. Halmahera Timur Stainles Steel	(250 MW)
21. Kariangau Oil & Gas	(397.,9 MW)
22. Batu Licin Steel	(500 MW)
23. Landak Rubber	(400 MW)
24. Tayan Alumina	(400 MW)
25. Bangka Tin	(153,08 MW)
26. Tanjung Api-Api Coal Gasi cation	(808,92 MW
27. Muara Enim Rubber	(120 MW)
28. Tanggamus Maritim	(700 MW)
29. Tanjung BUton Oleochemical	(1.000 MW)
30. Kuala Tanjung Alumina	(200 MW)
31. Sei Ba[illegible] Rubber	(22,5 MW)
32. Sei Mangke Palm Oil	(400,4 MW)

Total : 10,864,38 MW

Source: Ministry of Industry, reprocessed by NEC

ENERGY DEMAND FOR INDUSTRIAL SECTOR

(GWh)

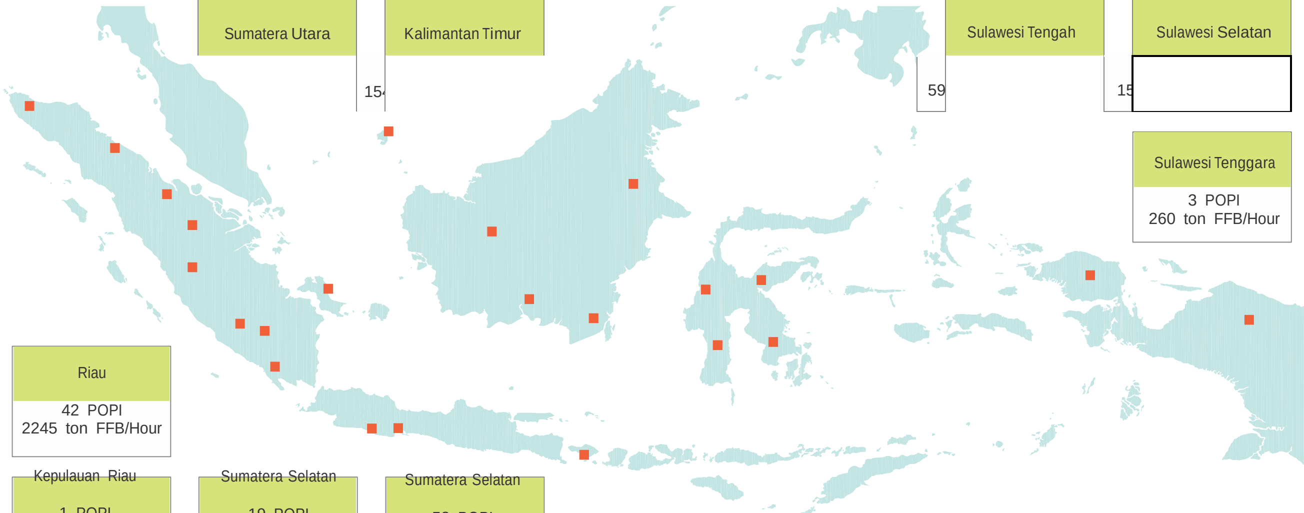


Source: Ministry of Industry and BPS, reprocessed by NEC

Beyond the need for feedstock in the fertilizer industry, the pulp and paper industry is the largest user of energy i.e. by 59.3 percent. The cement industry is projected to grow most rapidly, from 6.5 percent in 2012 to 12.4 percent in 2025

POWER PLANT POTENTIAL BASED ON PALM OIL WASTE

Province	POPI	Capacity
Nanggroe Aceh Darussalam	25 POPI	980 ton FFB/Hour
Sumatera Utara	92 POPI	3815 ton FFB/Hour
Sumatera Barat	26 POPI	1645 ton FFB/Hour
Kalimantan Barat	65 POPI	5475 ton FFB/Hour
Kalimantan Tengah	43 POPI	3100 ton FFB/Hour
Kalimantan Selatan	15 POPI	770 ton FFB/Hour
Sulawesi Barat	6 POPI	260 ton FFB/Hour
Sumatera Utara	15	
Kalimantan Timur		
Sulawesi Tengah	59	
Sulawesi Selatan	15	
Sulawesi Tenggara	3 POPI	260 ton FFB/Hour
Riau	42 POPI	2245 ton FFB/Hour
Kepulauan Riau	1 POPI	40 ton FFB/Hour
Sumatera Selatan	19 POPI	990 ton FFB/Hour
Sumatera Selatan	58 POPI	3555 ton FFB/Hour
Bangka Belitung	16 POPI	1235 ton FFB/Hour
Lampung	10 POPI	375 ton FFB/Hour
Jawa Barat	1 POPI	60 ton FFB/Hour
Papua	3 POPI	140 TPH
Papua Barat	4 POPI	360 ton FFB/Hour



Source : Ministry of Agriculture and BPS

CO_2 EMISSIONS

NATIONAL COMMITMENT TO REDUCE GHG

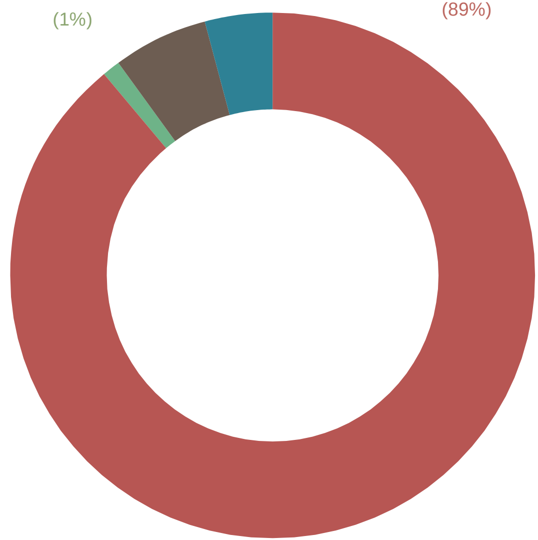
Presidential Regulation Number 61 Year 2011
to reduce greenhouse gasses (GHG) emission by 2020
(Indonesia Commitment in G-20 Pittsburgh and COP15)

26%
(0.767 Giga Tonne CO2e)
Domestic e ort

41%
(1.189 Giga Tonne CO2e)
Domestic e ort and international support

New Energy Development

Energy Conservation



BP Statistical Review, reprocessed by NEC

EMISSION REDUCTION ROAD MAP IN ENERGY SECTOR

ACTION PLAN

Million Ton CO2

2010-2014

5.06

- Energy Audit
- Energy Saving Lamps
- The provision and management of NRE
- Biogas utilization
- Gas for public transportation
- Increasing gas city connection to households
- Developing the LPG Mini Refinery Plant
- Reclamation for the ex-mining location

2015-2020

18.08

- Energy Audit
- Energy Saving Lamps
- The provision and management of NRE
- Biogas utilization
- Gas for public transportation
- Increasing gas city connection to households
- Developing the LPG Mini Refinery Plant
- Reclamation for the ex-mining location

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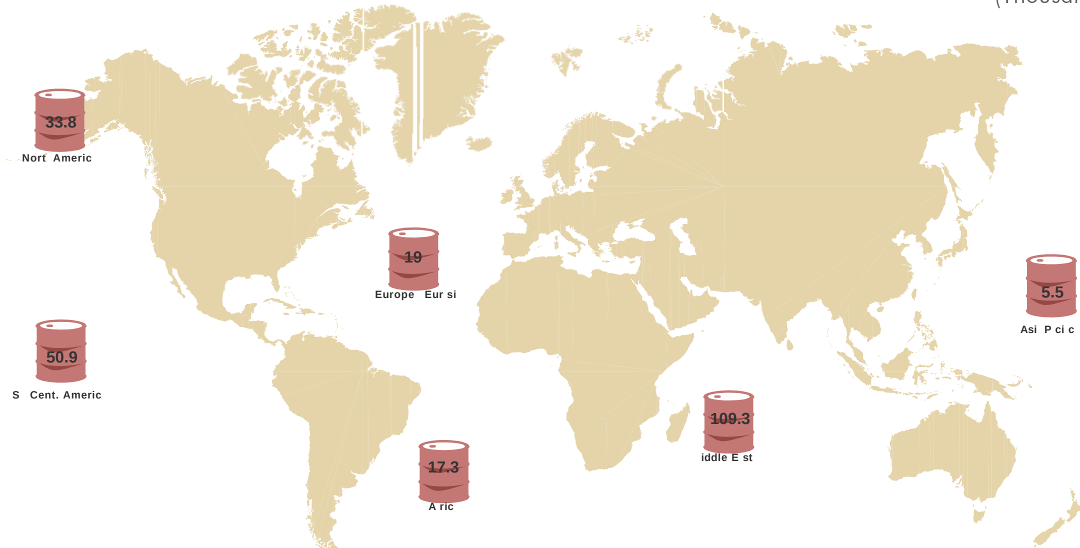
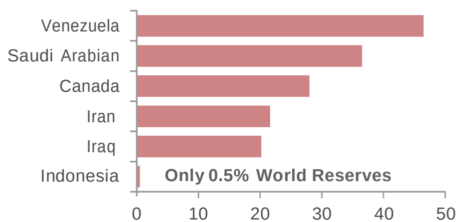
INDONESIA'S POSITION IN THE GLOBAL CONTEXT

OIL

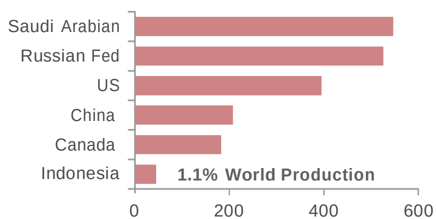
(Thousand Million Tonnes)



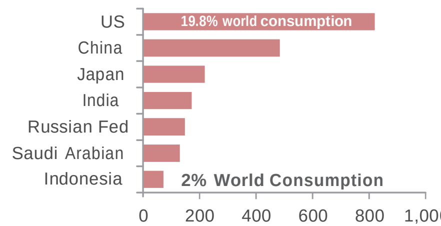
Proven Oil in selected countries 2012



Production Oil in selected countries 2012 T



Consumption Oil in selected countries 2012 T

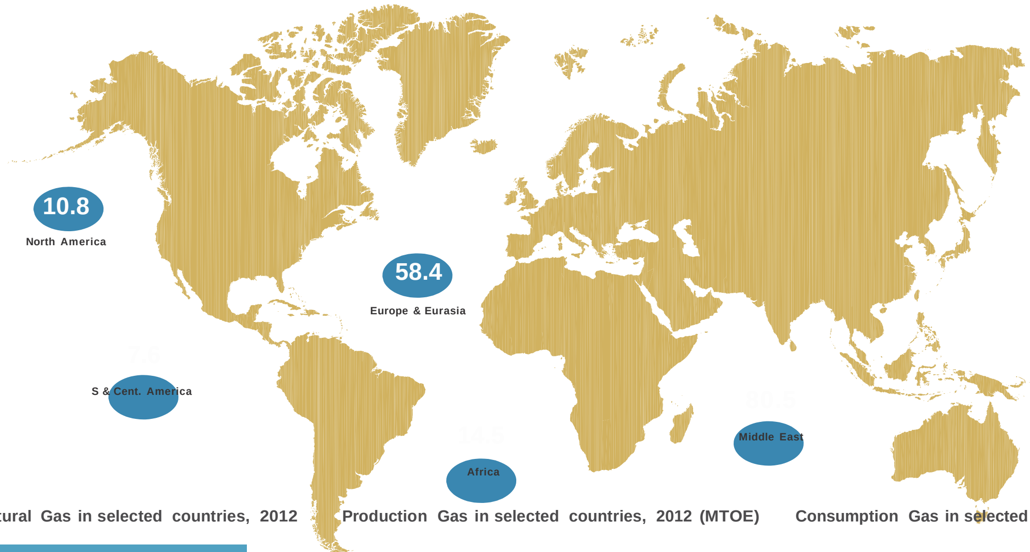


BP Statistical Review, reprocessed by NEC

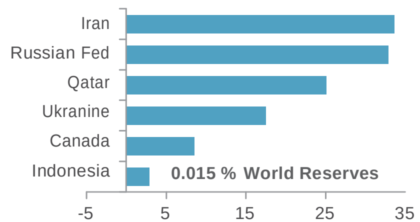
BP Statistical Review, reprocessed by NEC

GAS

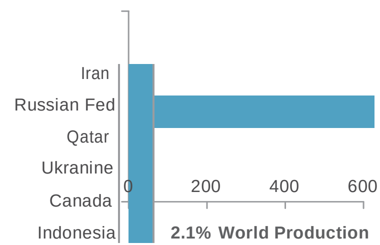
(TCM)



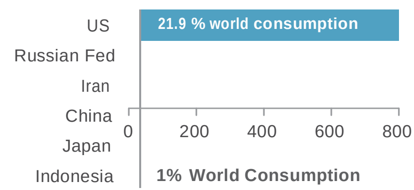
Proven Natural Gas in selected countries, 2012



Production Gas in selected countries, 2012 (MTOE)



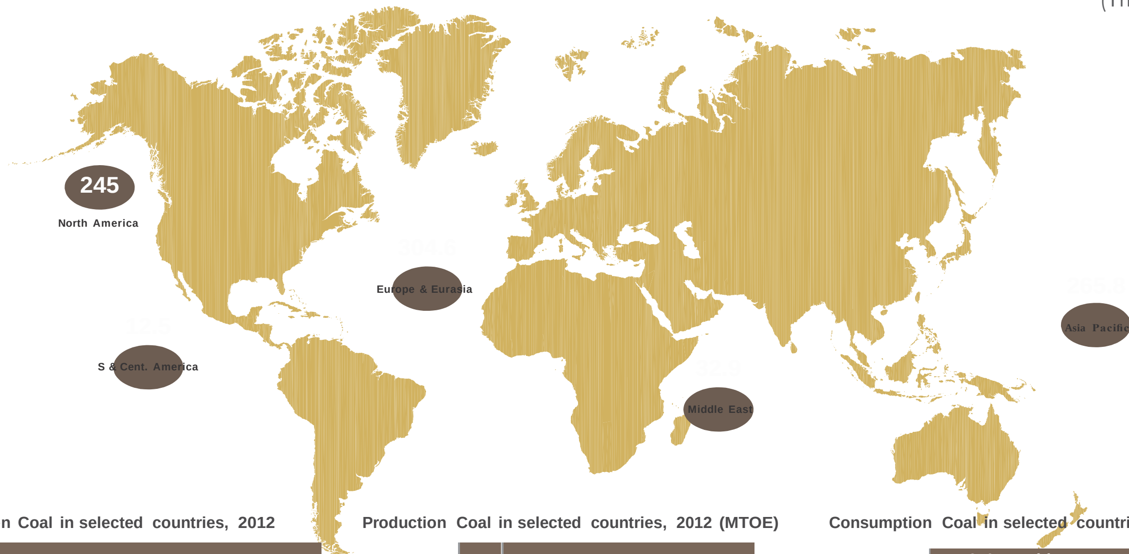
Consumption Gas in selected countries, 2012 (MTOE)



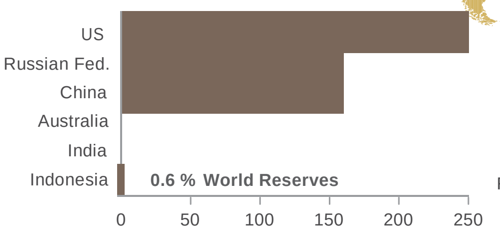
BP Statistical Review, reprocessed by NEC

COAL

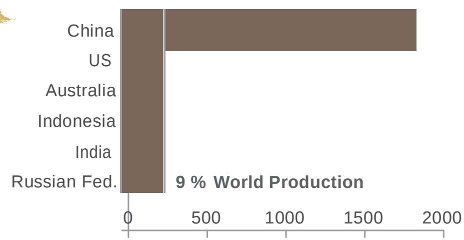
(Thousand MT)



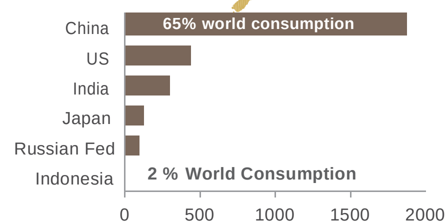
Proven Coal in selected countries, 2012



Production Coal in selected countries, 2012 (MTOE)

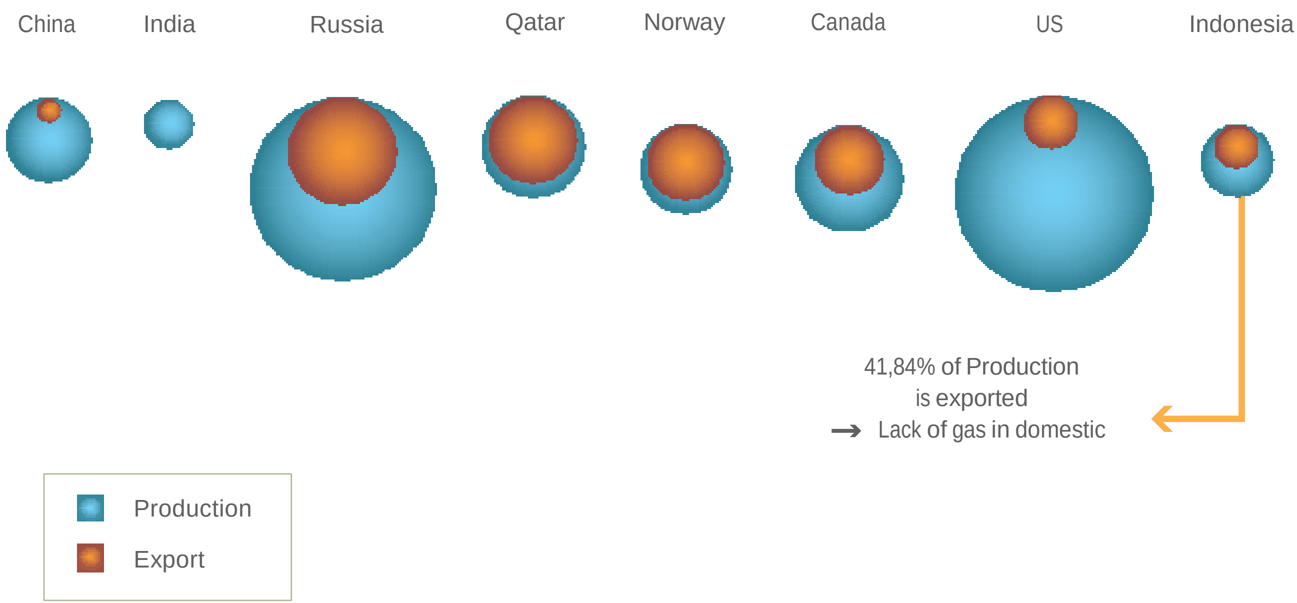


Consumption Coal in selected countries, 2012 (MTOE)



BP Statistical Review, reprocessed by NEC

GAS PRODUCTION-EXPORT COMPARISON



BP Statistical Review, reprocessed by NEC

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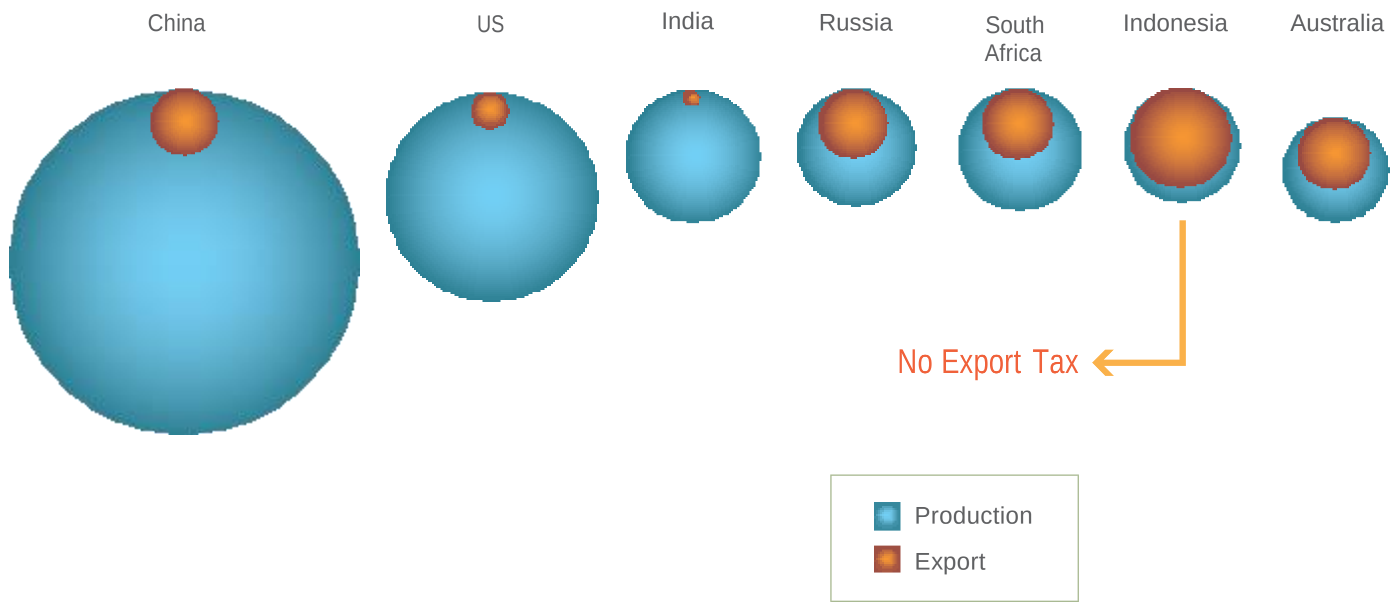
COAL PRODUCTION-EXPORT COMPARISON



Source: Wood Mackenziecoal supply service, ANZ, reprocessed by NEC

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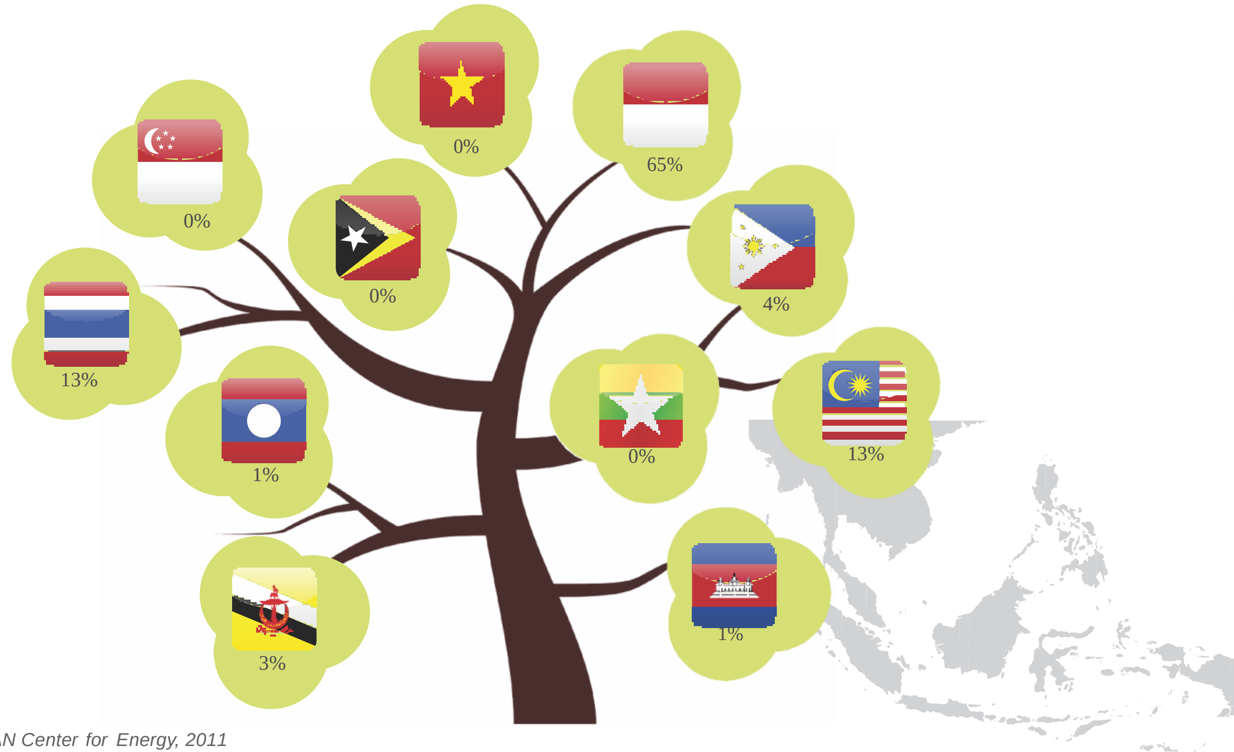
SHARE OF ASEAN PRIMARY ENERGY PRODUCTION



Source: ASEAN Center for Energy, 2011

Note : Data of Myanmar and Vietnam are not yet provided

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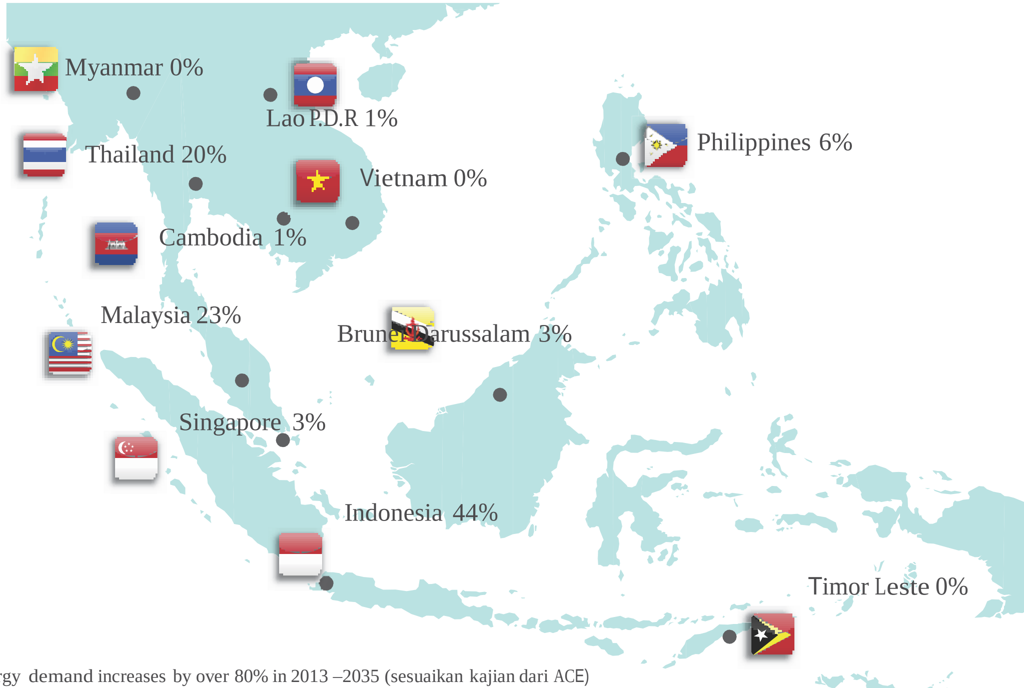
SHARE OF ASEAN ENERGY DEMAND

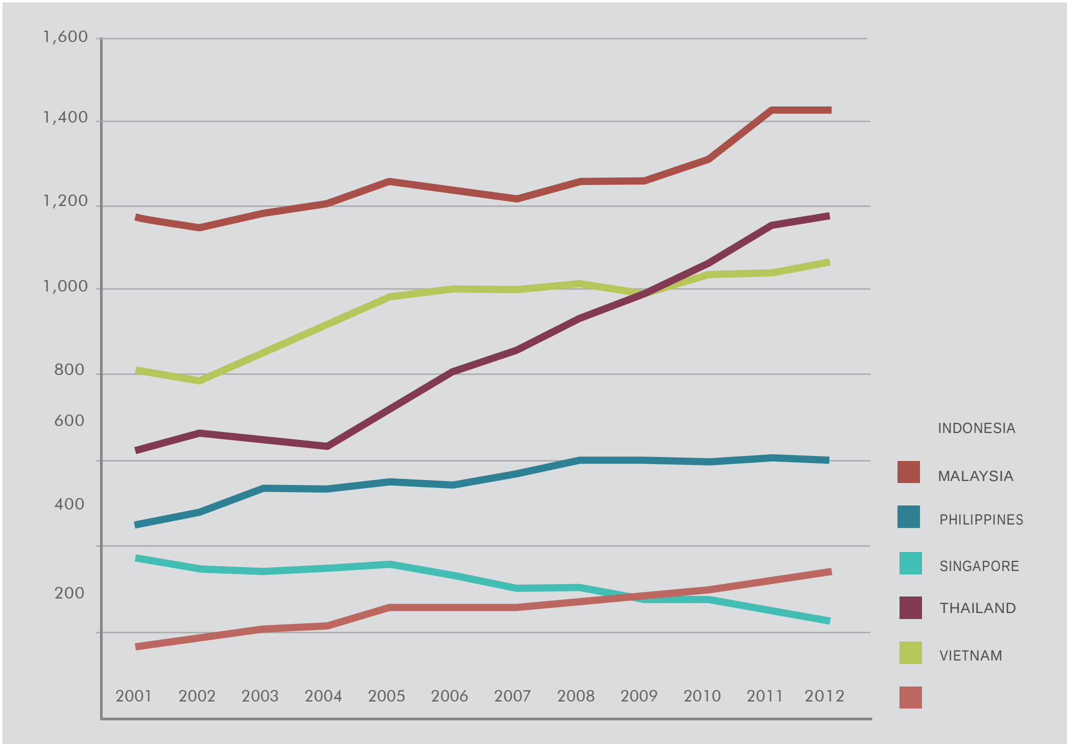


- Southeast Asia's energy demand increases by over 80% in 2013 –2035 (sesuaikan kajian dari ACE)
- Oil demand rises from 4.4 mb/d today to 6.8 mb/d in 2035, natural gas demand increases by 80% to 250 bcm

Source : ASEAN Center for Energy, 2011
Note : Data of Myanmar and Vietnam are not yet provided

ASEAN OIL CONSUMPTION

(Thousand Barrel per Day)



Source: Data of other ASEAN member states are not yet provided

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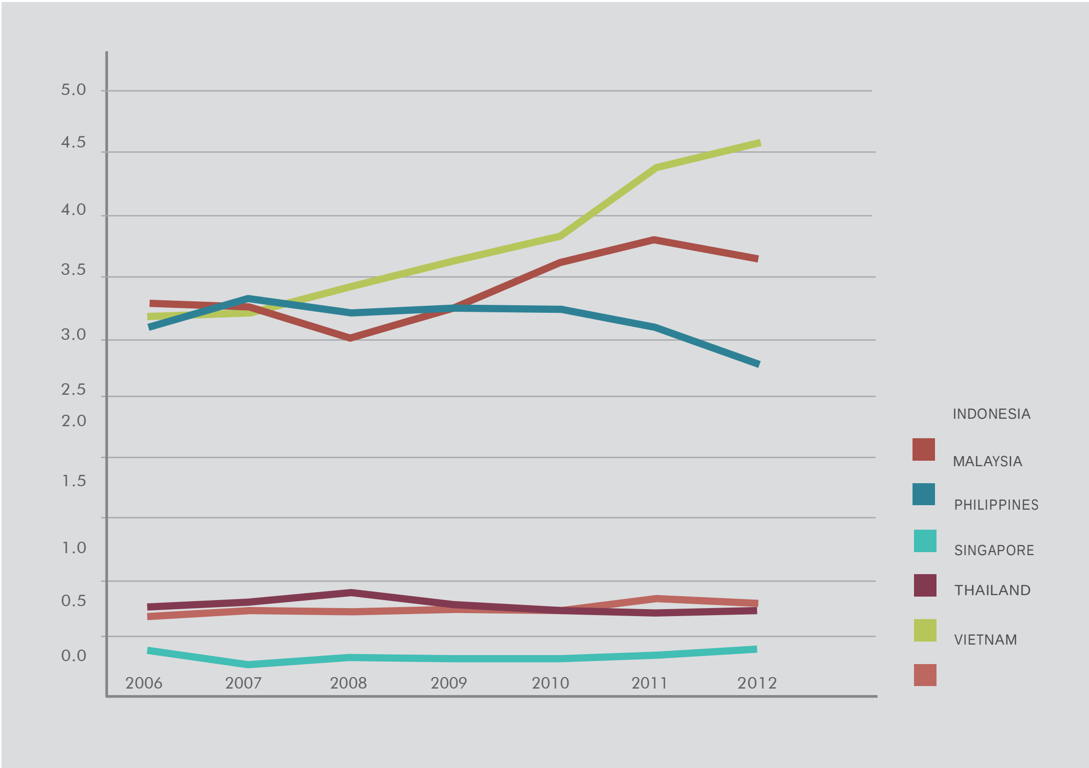
ASEAN GAS CONSUMPTION

(Thousand Barrel per Day)



Source: Data of other ASEAN member states are not yet provided

ABBREVIATION

Abbreviation	
Bbl	Barrel
BOE	Barrels of Oil Equivalent
BOEPD	Barrels Of Oil Equivalent Per Day
GDP	Gross Domestic Product
GW	Giga Watt
GWh	Giga Watt Hour
KL	Kilo Litter
LPG	Lique ed Petroleum Gas
MBOE	Million Barrels of Oil Equivalent
MMSTB	Million Stock Tank Barrels
MMSCF	Million Standard Cubic Feet
MT	Million Ton
MTOE	Million Ton of Oil Equivalent
MW	Mega Watt
MWh	Mega Watt Hour
KL	Kilo Litter
KWh	Kilo Watt Hour

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